Peer-to-peer protocols for efficient and resilient decentralized learning

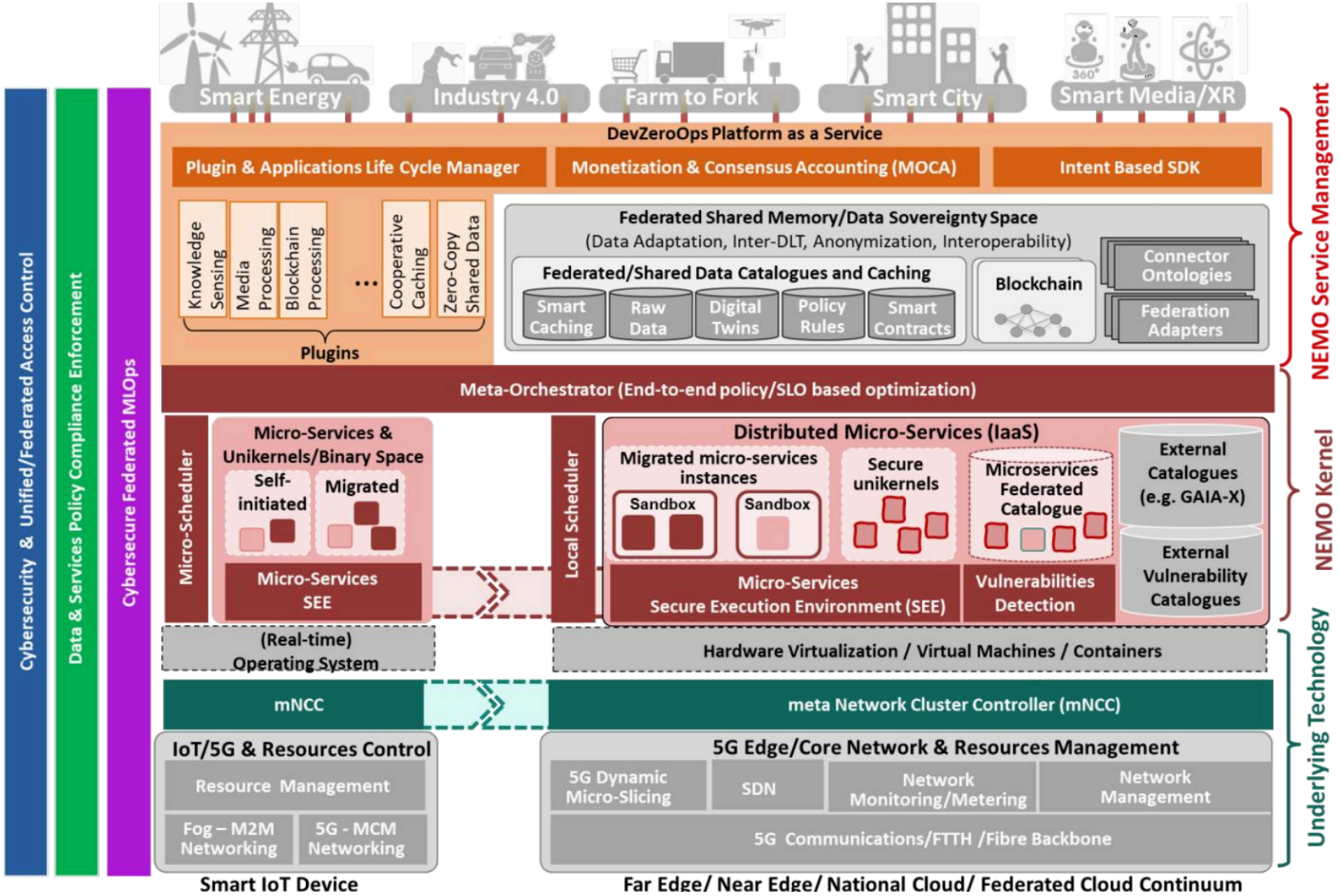
Mohamed Amine Legheraba

Supervisors: Maria Potop-Butucaru, Sébastien Tixeuil

LIP6 · Sorbonne Université

September 7, 2026

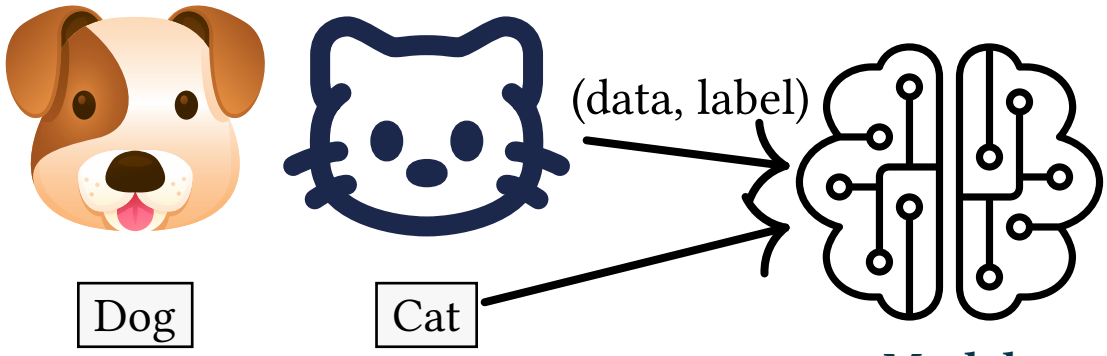
Context


Smart Energy
Industry 4.0
Farm to Fork
Smart City
Smart Media/XR
Cybersecurity & Unified/Federated Access Control
Data & Services Policy Compliance Enforcement
Cybersecure Federated MLOps
DevZeroOps Platform as a Service
Plugin & Applications Life Cycle Manager
Monetization & Consensus Accounting (MOCA)
Intent Based SDK
Knowledge Sensing
Media Processing
Blockchain Processing
...
Cooperative Caching
Zero-Copy Shared Data
Plugins
Federated Shared Memory/Data Sovereignty Space
(Data Adaptation, Inter-DLT, Anonymization, Interoperability)
Federated/Shared Data Catalogues and Caching
Smart Caching
Raw Data
Digital Twins
Policy Rules
Smart Contracts
Blockchain
Connector Ontologies
Federation Adapters
NEMO Service Management
Meta-Orchestrator (End-to-end policy/SLO based optimization)
Micro-Scheduler
Micro-Services & Unikernels/Binary Space
Self-initiated
Migrated
Micro-Services SEE
Local Scheduler
Distributed Micro-Services (IaaS)
Migrated micro-services instances
Sandbox
Sandbox
Secure unikernels
Microservices Federated Catalogue
External Catalogues (e.g. GAIA-X)
External Vulnerability Catalogues
Micro-Services Secure Execution Environment (SEE)
Vulnerabilities Detection
NEMO Kernel
(Real-time) Operating System
Hardware Virtualization / Virtual Machines / Containers
mNCC
meta Network Cluster Controller (mNCC)
IoT/5G & Resources Control
Resource Management
Fog – M2M Networking
5G - MCM Networking
5G Edge/Core Network & Resources Management
5G Dynamic Micro-Slicing
SDN
Network Monitoring/Metering
Network Management
5G Communications/FTTH /Fibre Backbone
Underlying Technology
Smart IoT Device
Far Edge/ Near Edge/ National Cloud/ Federated Cloud Continuum

Supervised Classification

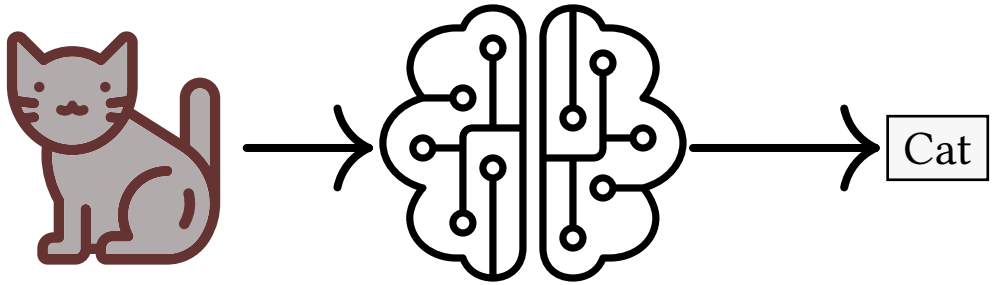
Training

Data

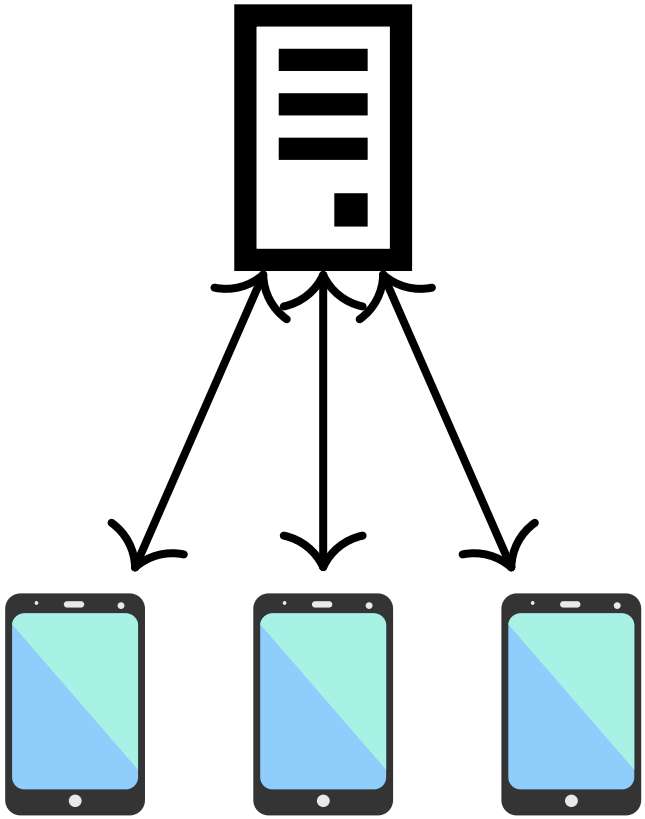


Labels

Prediction



Federated Learning



Server

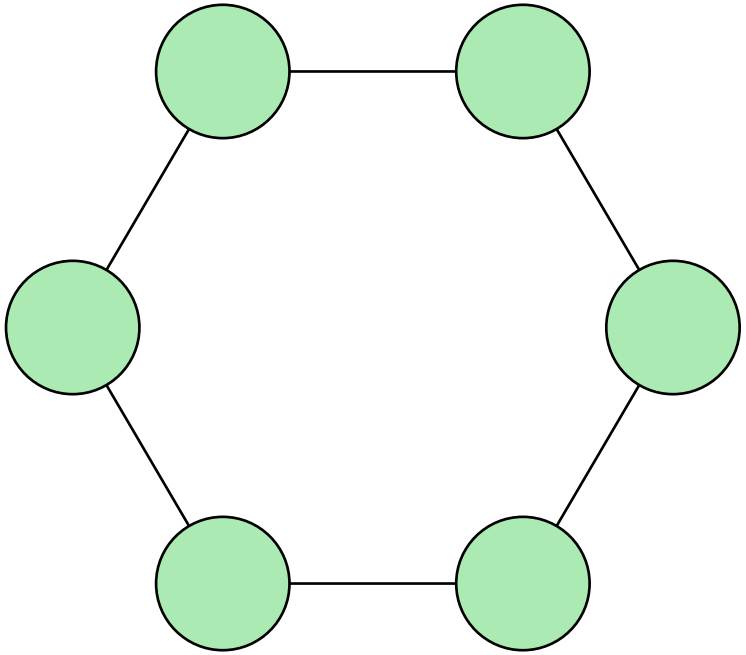
```
loop
    received_models ← collect(clients)
    global_model ← merge(received_models)
    broadcast(global_model)
```

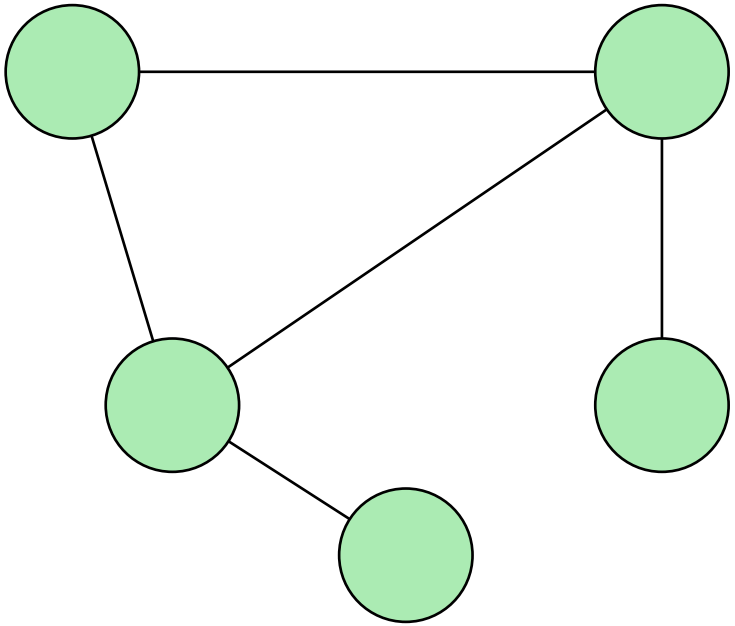
Client

```
loop
    local_model ← train(model, local_data)
    send(local_model, server)
    model ← receive(global_model)
```

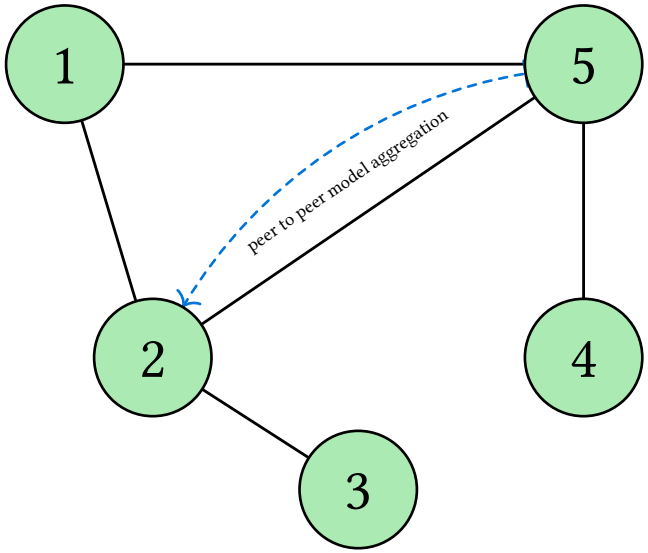
[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*, 2017, pp. 1273–1282.

Structured vs Unstructured networks





Gossip Learning

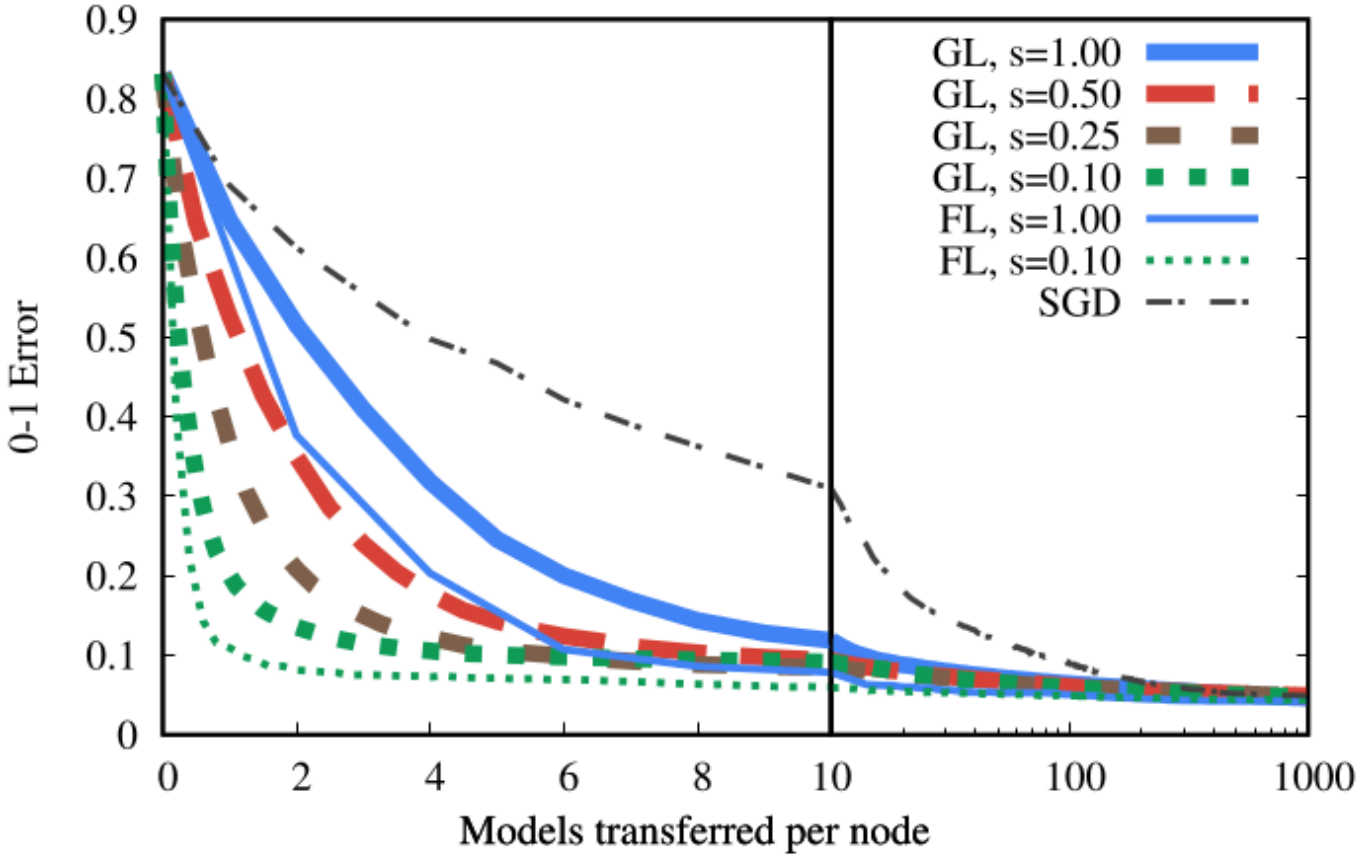


Gossip Learning (main thread)

ML model: **model**
List of neighbors : **cache**
1 **loop**
2 Wait for **Δ** time units
3 peer ← selectRandom(cache)
4 send(peer, model)

[2] R. Ormándi, I. Hegedűs, and M. Jelasity, "Gossip learning with linear models on fully distributed data," *Concurrency and Computation: Practice and Experience*, vol. 25, no. 4, pp. 556–571, 2013.

Gossip Learning vs Federated Learning



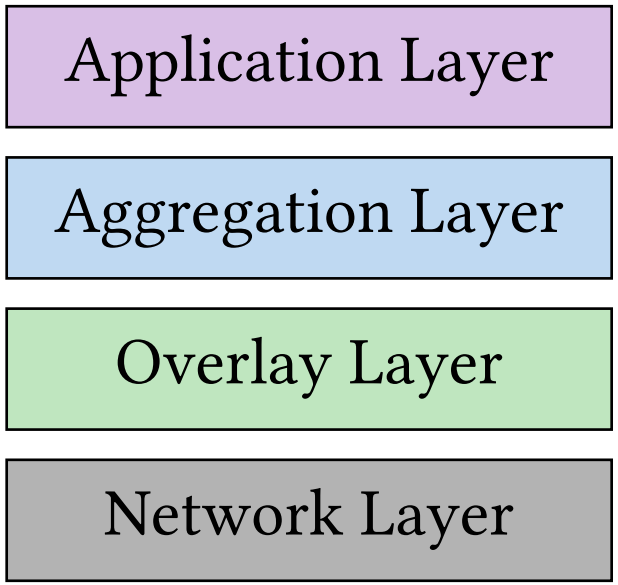
[3] I. Hegedűs, G. Danner, and M. Jelasity, "Decentralized learning works: An empirical comparison of gossip learning and federated learning," *Journal of Parallel and Distributed Computing*, vol. 148, pp. 109–124, 2021.

State of the Art Summary

	Decentralized	Local data	Fast convergence	Fault-tolerant	No overlay overhead
Centralized learning	✗	✗	✓	✗	✓
Federated learning	✗	✓	✓	✗	✓
Decentralized learning (structured net.)	✓	✓	✓	✗	✗
Gossip learning	✓	✓	✗	✓	✓
Local learning	✓	✓	✗ no generalizing model	✓	✓

No single solution is fully satisfactory

Contributions



Outline

- **Elevator** — overlay (hub election)
- **Lift** — malicious resilience
- **HEAL** — model aggregation
- **FLAIR** — wireless networks adaptation

Elevator

Hub-based topology

Definition. (Hub) Let $G = (V, E)$ be the directed overlay graph, where each node $v \in V$ keeps a partial view $C(v) \subseteq V$.

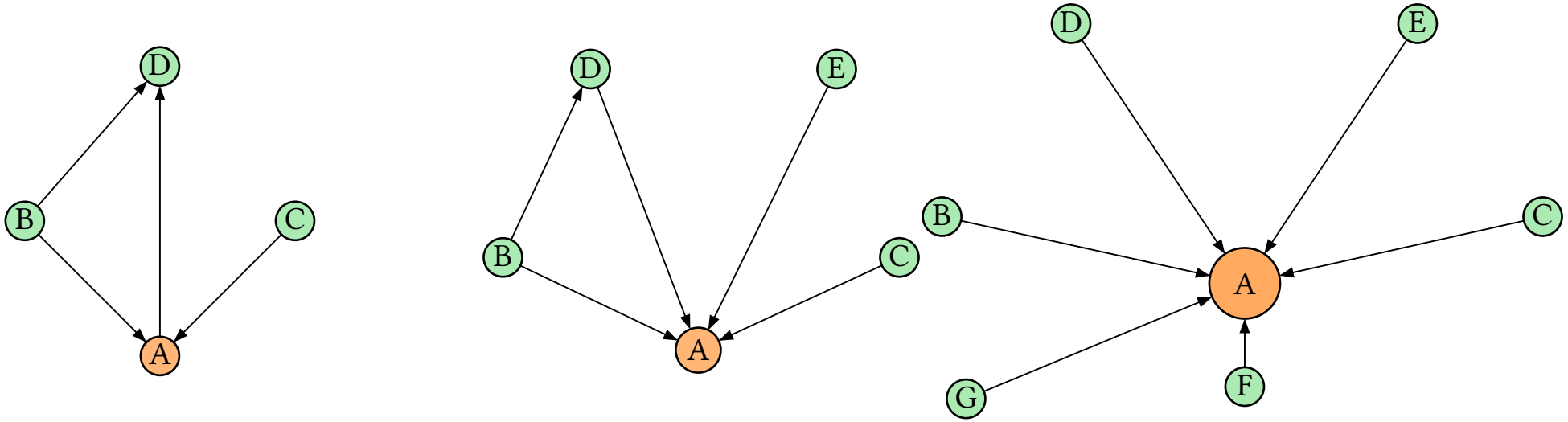
A node $h \in V$ is a **hub** if it appears in the partial view of every node:

$$\forall v \in V, \quad h \in C(v)$$

[4] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, "Emergent Peer-to-Peer Multi-Hub Topology," in *2024 22nd International Symposium on Network Computing and Applications (NCA)*, 2024, pp. 219–226.

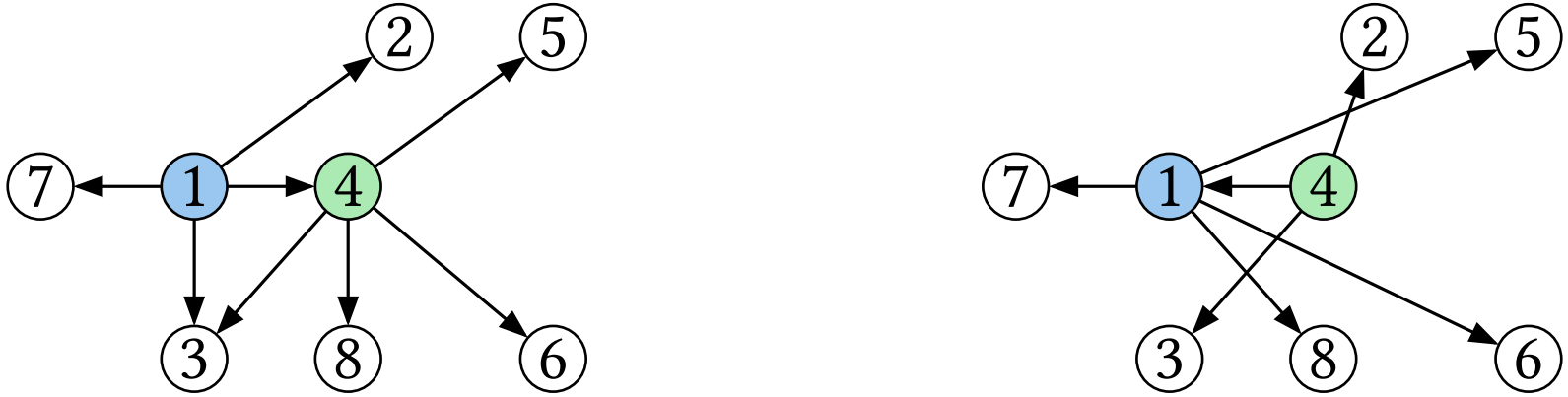
Preferential Attachment

1. **One node is more connected**
2. **It attracts more connections**
3. **Snowball effect: it becomes a hub**



[5] A.-L. Barabâsi, H. Jeong, Z. Néda, E. Ravasz, A. Schubert, and T. Vicsek, "Evolution of the social network of scientific collaborations," *Physica A: Statistical mechanics and its applications*, vol. 311, no. 3–4, pp. 590–614, 2002.

Random Attachment



Before and after a shuffling operation. Node 1 sends addresses {itself, 2, 3} to node 4. Node 4 sends back {5,6, 8}.

[6] A. Stavrou, D. Rubenstein, and S. Sahu, "A lightweight, robust p2p system to handle flash crowds," in *10th IEEE International Conference on Network Protocols, 2002. Proceedings.*, 2002, pp. 226–235.

Combining both ideas?

Preferential attachment makes hubs emerge…

Random attachment keeps the network resilient…

Can we combine both to get hubs that are resilient?

Elevator in a nutshell

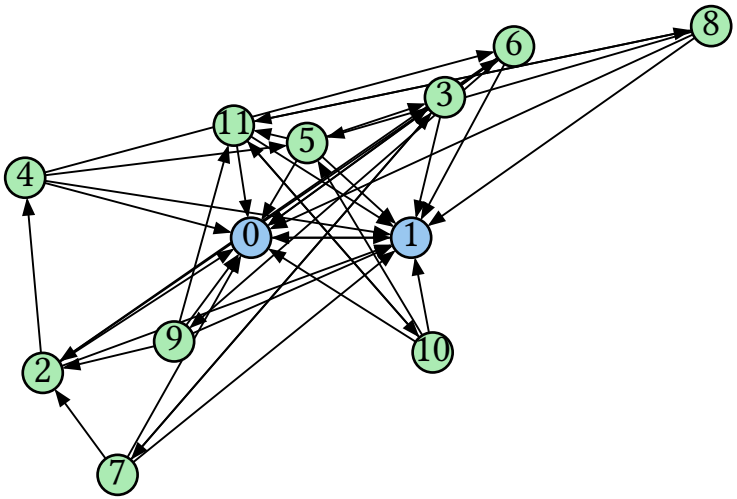
Mechanism

1. Each node asks its **neighbours** for **their** neighbour lists.
2. It connects to the ***h* most frequent** nodes.
3. It asks these h hubs for extra incoming connections, to fill a **random subset** of its view.

Observed

- After a few cycles, **hubs emerge** (the preferred h are the same for everyone);
- Hubs are **elected at random** among the network nodes;
- Even after **hub failures**, new nodes are elected as hubs.

Example of topology generated



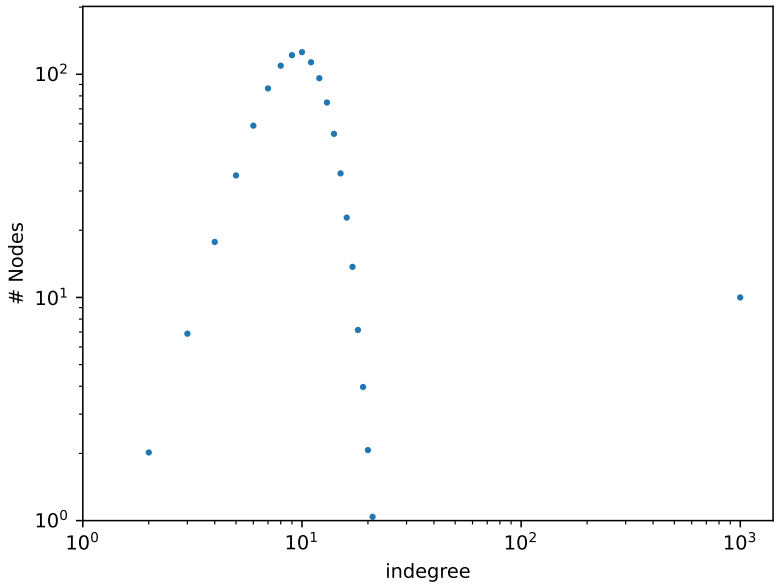


Figure 1: Indegree distribution of a network generated with Elevator, with 1000 nodes and 10 hubs

Stability

Definition. (Stability) Once Elevator has converged to a set of h hubs, both the list of hubs and their number h remain constant (with high probability) over time.

Formally, after convergence time T:

$$\Pr[\forall t \geq T, H(t) = H(T) \wedge \forall v \in V, C_v(t)[1:h] = H(t)] = 1$$

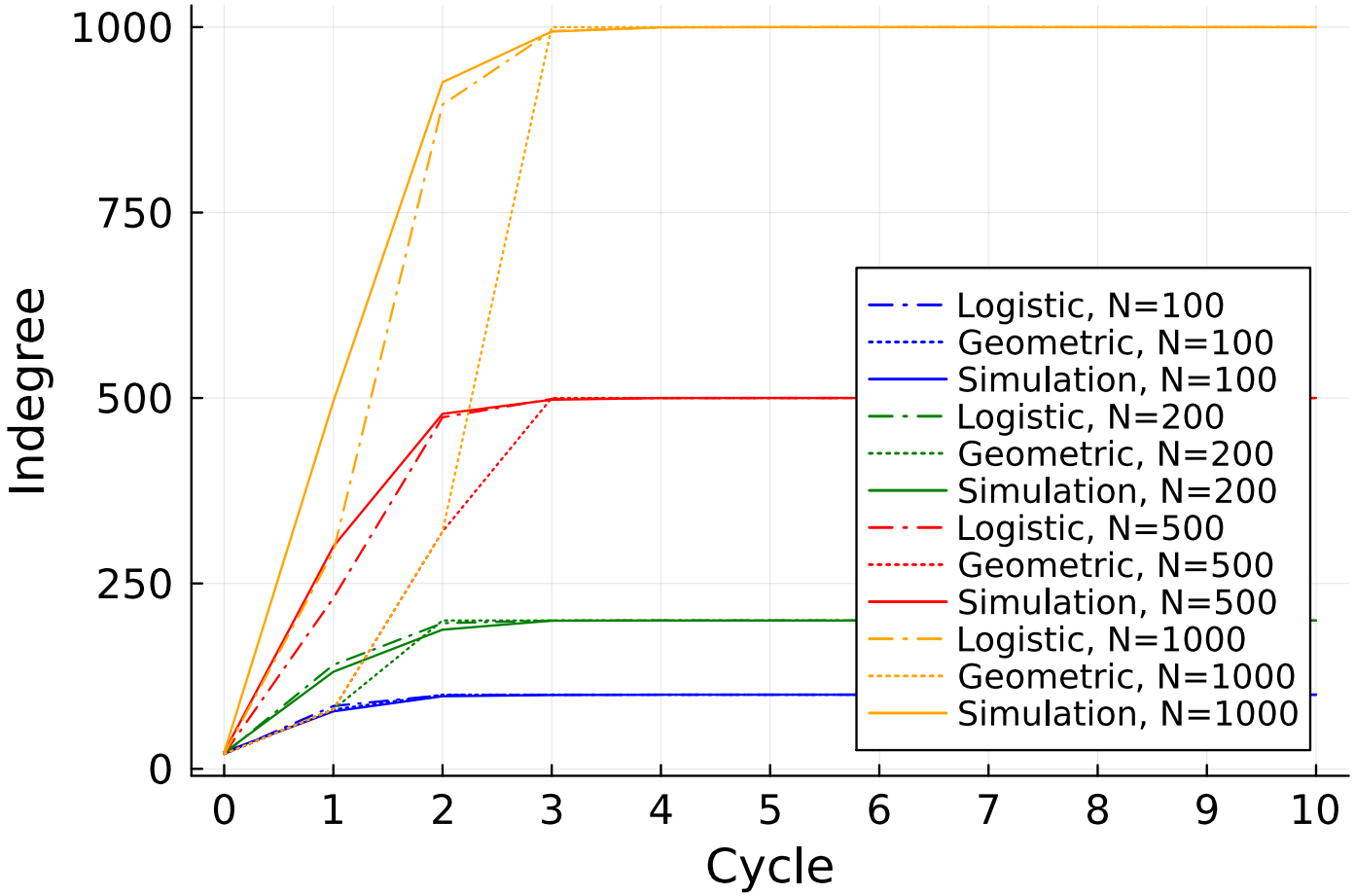
Convergence

Definition. (Convergence) The network converges to a stable state containing **exactly** h hubs shared by all nodes.

Idea of the proof.

- A new hub is produced with **non-zero probability** at each cycle.
- The number of hubs **cannot decrease** (until it reaches h).
- Once at least one hub exists, the network is **strongly connected** (w.h.p.) and produces a new hub.

Speed of convergence



Simulations

PeerSim was **forked and substantially rewritten** for this thesis:

- Migrated from **SVN to Git**, build rewritten in **Gradle**;
- **Docker** + **GitLab CI/CD**;
- **Parallelised** the cycle-based engine;
- Failure models from scratch (**crash, churn, Byzantine**);

Parameters

- Network size: **N = 1000 nodes**;
- **1000 cycles**, repeated **100 times**;
- Initial topology: **random k-out graph** with **k = c = 20**;
- **h = 10** hubs;
- e.g. brutal crash: **50% of nodes** disconnected at cycle **500**.

Metrics

Overlay

- In-degree distribution
- Clustering coefficient
- Average path length
- Diameter
- Convergence time

Machine learning

- Accuracy

Failures

Overlay

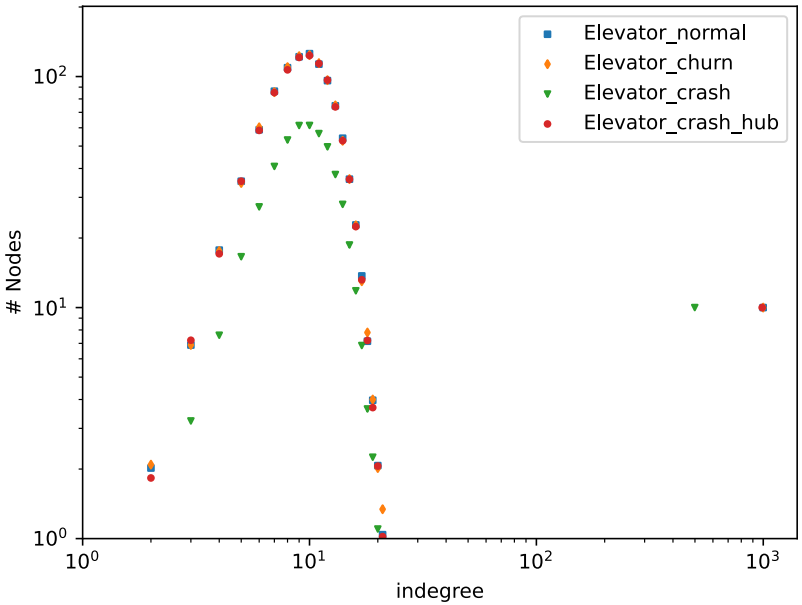
- **Crash failures**
- **Churn** (dynamic joins / leaves)
- **Byzantine failures**

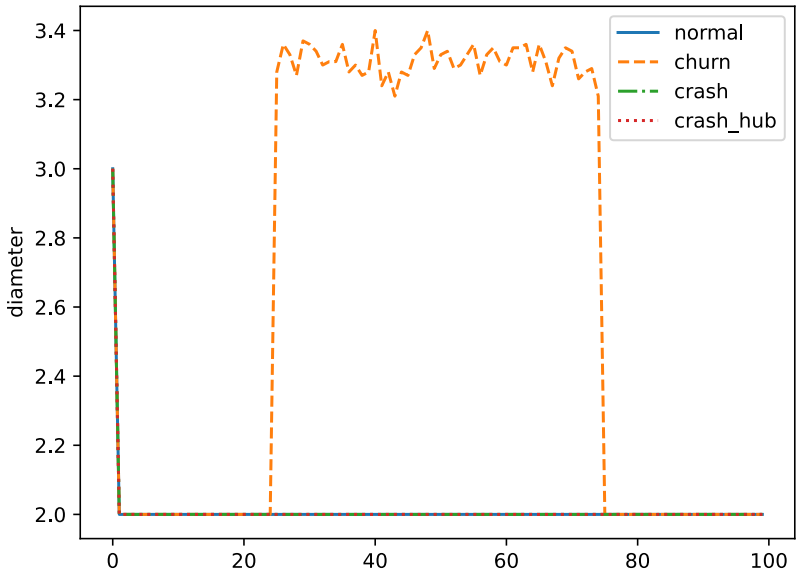
Machine learning

- **Privacy attacks**
- **Poisoning attacks**

Note: ML-level failures are not addressed in this thesis

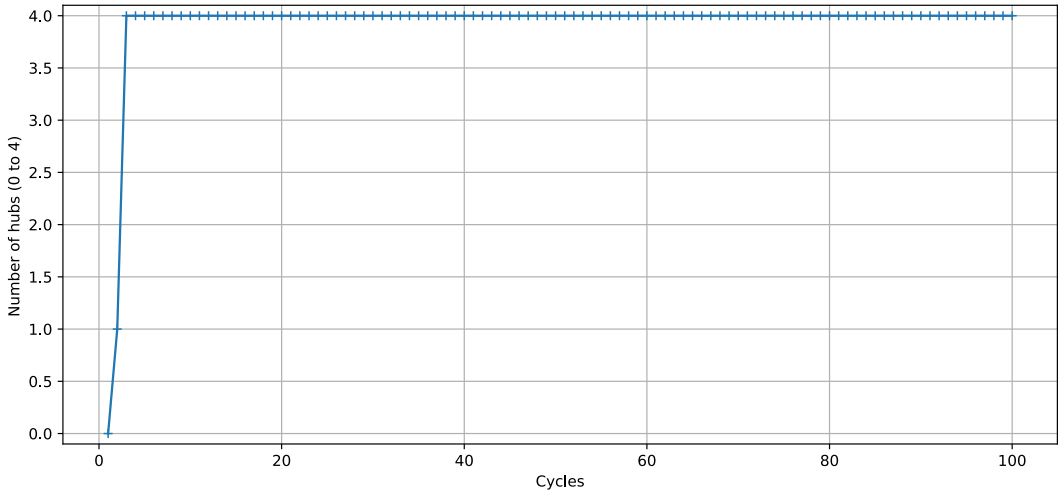
Resilience


Elevator_normal
Elevator_churn
Elevator_crash
Elevator_crash_hub
Nodes
indegree
10^0
10^1
10^2
10^3


normal
churn
crash
crash_hub
diameter
3.4
3.2
3.0
2.8
2.6
2.4
2.2
2.0
0
20
40
60
80
100

Real TCP/IP experiments

- **Go + libp2p** over **TCP/IP streams** (+ light **HTTP** per node);
- networks of **20 to 50 nodes** on **1–2 machines**;
- modes: **synchronous / externally-synchronized / asynchronous**;
- **hubs emerge within the first few cycles** — matching theory & simulation.



Lift

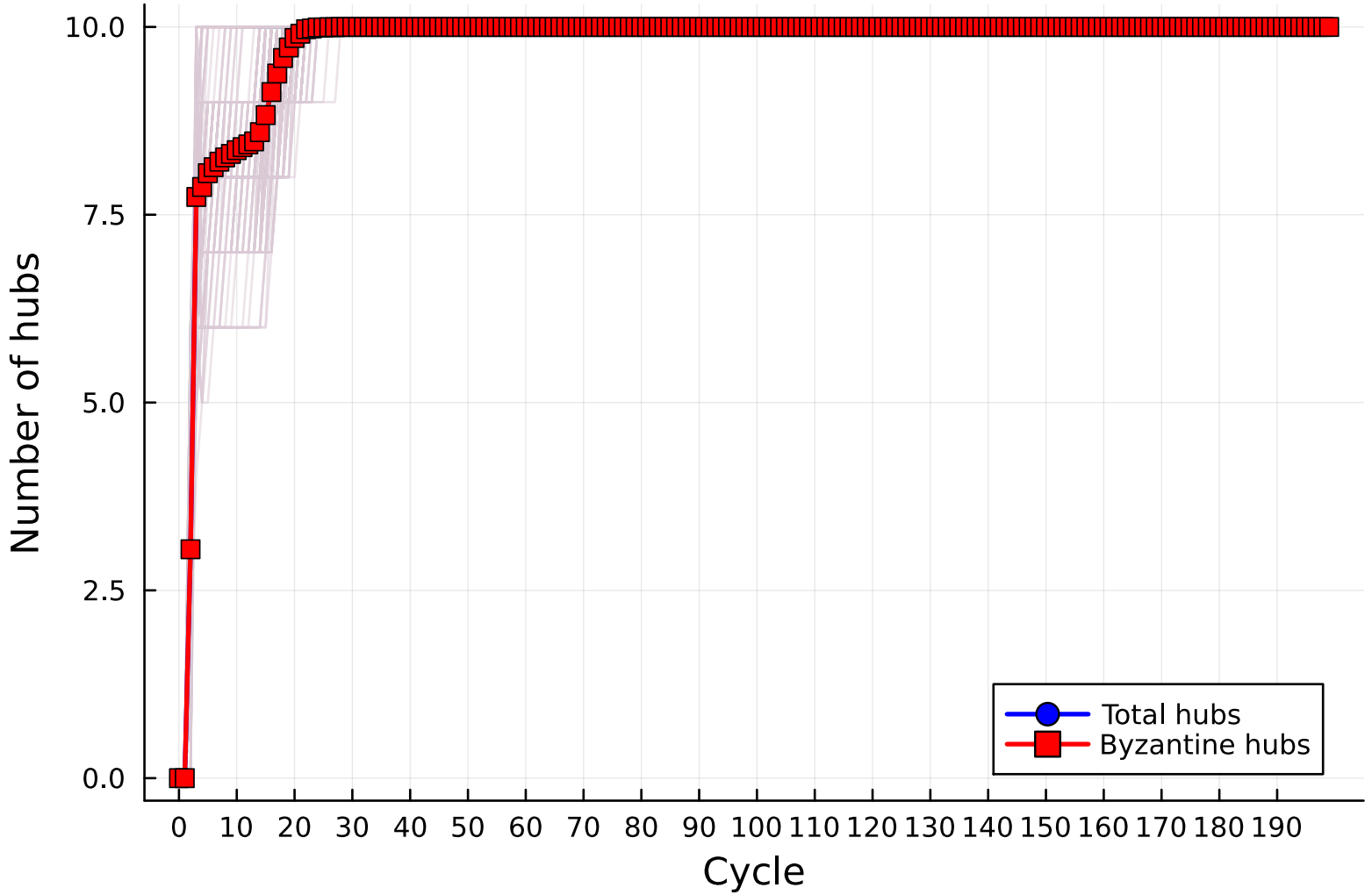
Malicious nodes

Coordinated attack — lying about neighbors

```
   addresses of all colluding nodes: colluders
   backward list: backward_peers
1  loop
2      (request, peer) ← receive()
3      if request == CACHE_REQUEST
4          backward_peers.add(peer)
5          colluders.shuffle()
6          modified_cache ← colluders[0:c]
7          send(modified_cache, peer)
```

Each colluding node answers with a fake cache that only contains **other colluding nodes**, inflating their visibility.

Attack outcome



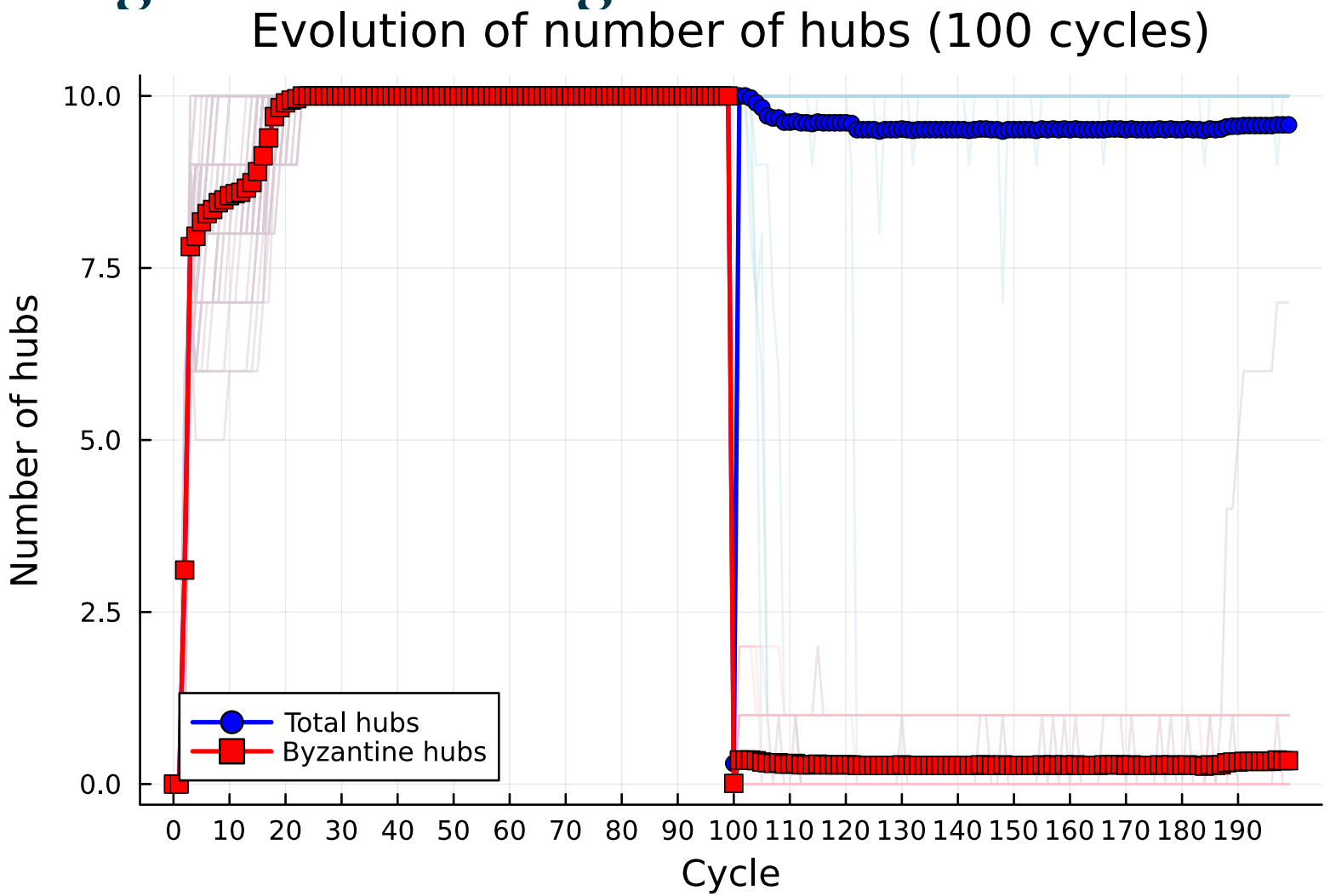
After convergence, correct nodes **deterministically** re-draw the hubs from a shared random seed — attackers cannot bias the outcome.

Hub redistribution

current hub list: **H**
network size: **N**
target hubs: **h**

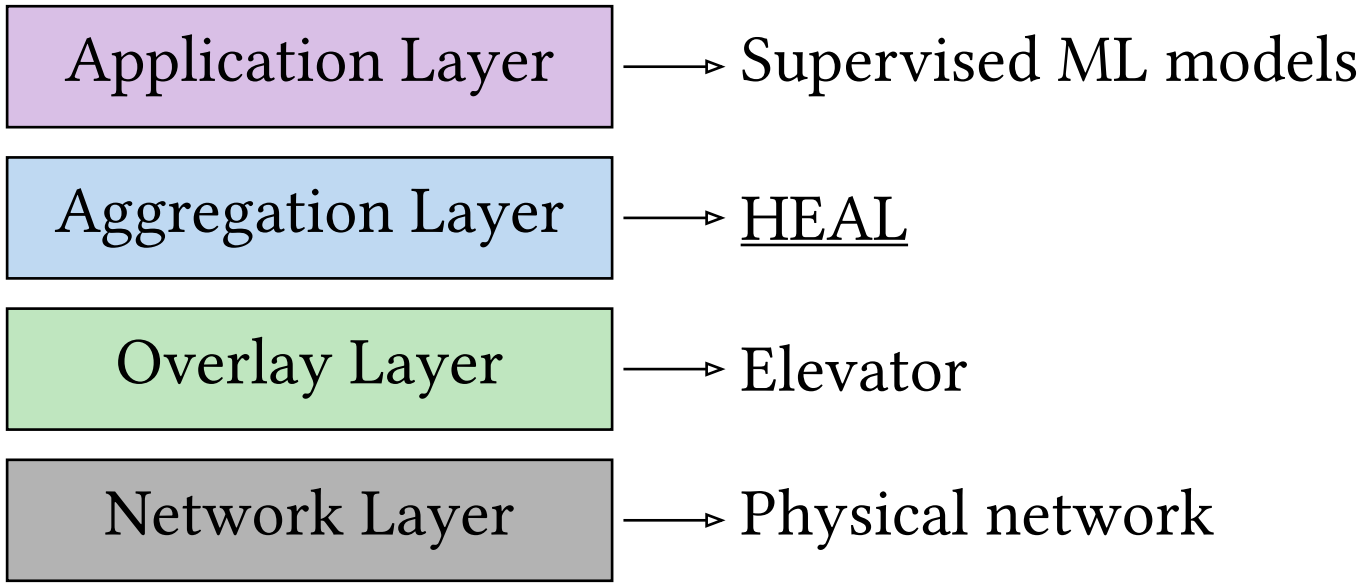
```
1 seed ← sort(H)
2 prng ← Random(seed)
3 selected ← {}
4 while selected.size() < h
5 |   id ← prng.nextInt(N)
6 |   if id not in selected
7 |   |   selected ← selected ∪ {id}
8 H ← selected
```

Resilience against colluding attackers



HEAL

Architecture

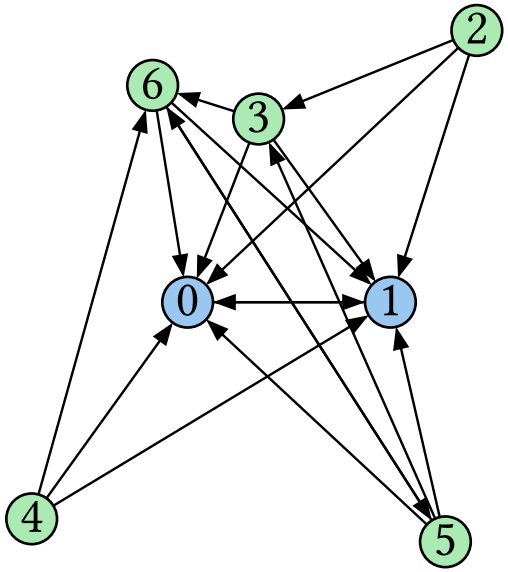


Hub Learning Protocol

SORBONNE UNIVERSITÉ

1) Each node train a local model

2) Each node sends its model to a (random) hub



3) Hubs receive and aggregate models

4) Hubs compute a global model together

5) Hubs send back the global model to the nodes

[7] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeuil, "HEAL: Resilient and Self-* Hub-based Learning," in *International Conference on Advanced Information Networking and Applications*, 2025, pp. 200–211.

Datasets & Models

Spambase + Logistic regression

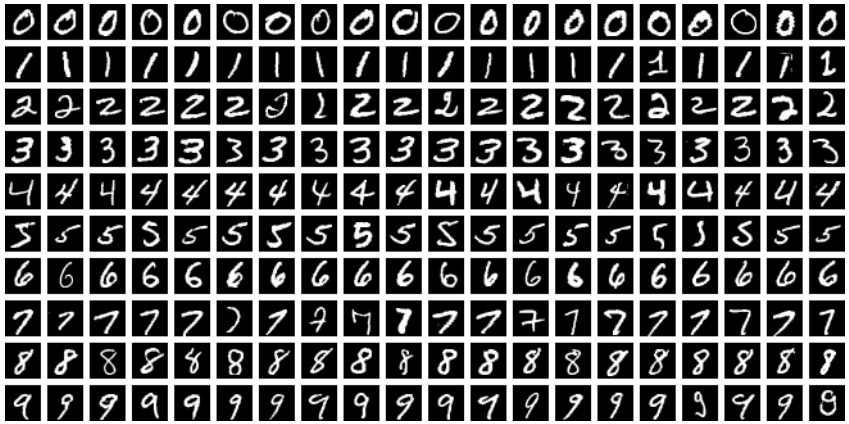
Variables Table

Variable Name	Role	Type
word_freq_make	Feature	Continuous
word_freq_address	Feature	Continuous
word_freq_all	Feature	Continuous
word_freq_3d	Feature	Continuous
word_freq_our	Feature	Continuous
word_freq_over	Feature	Continuous
word_freq_remove	Feature	Continuous

4601 emails, binary

57 parameters

MNIST + LeNet5

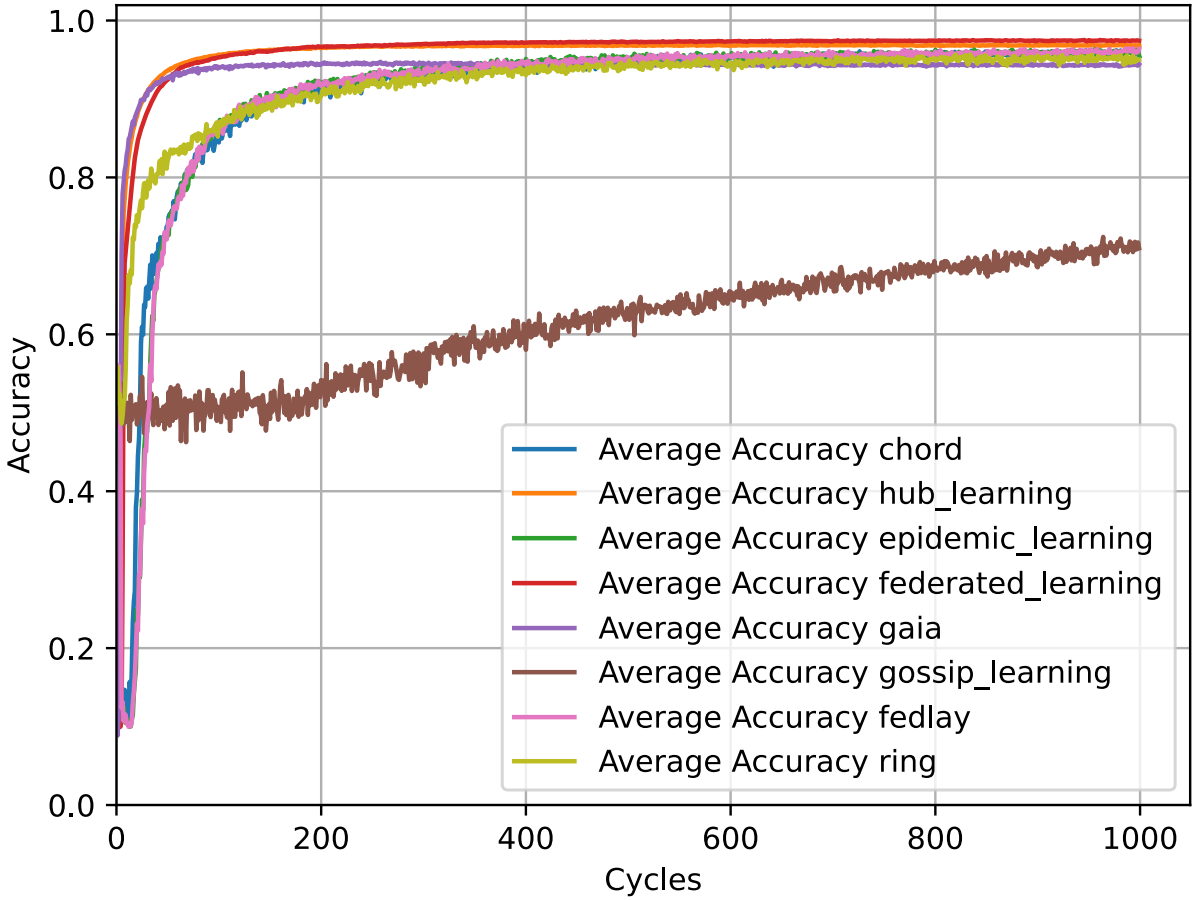


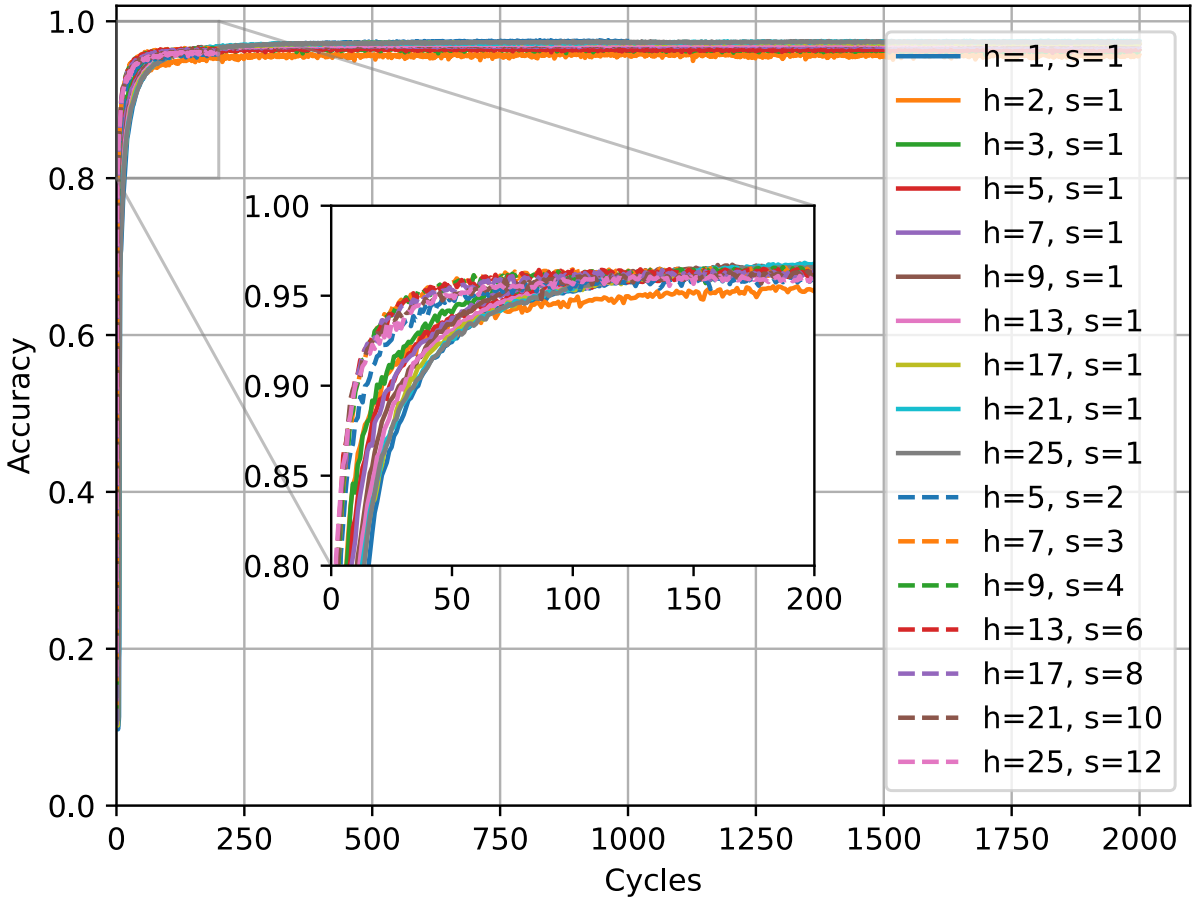
70k digits, 10 classes

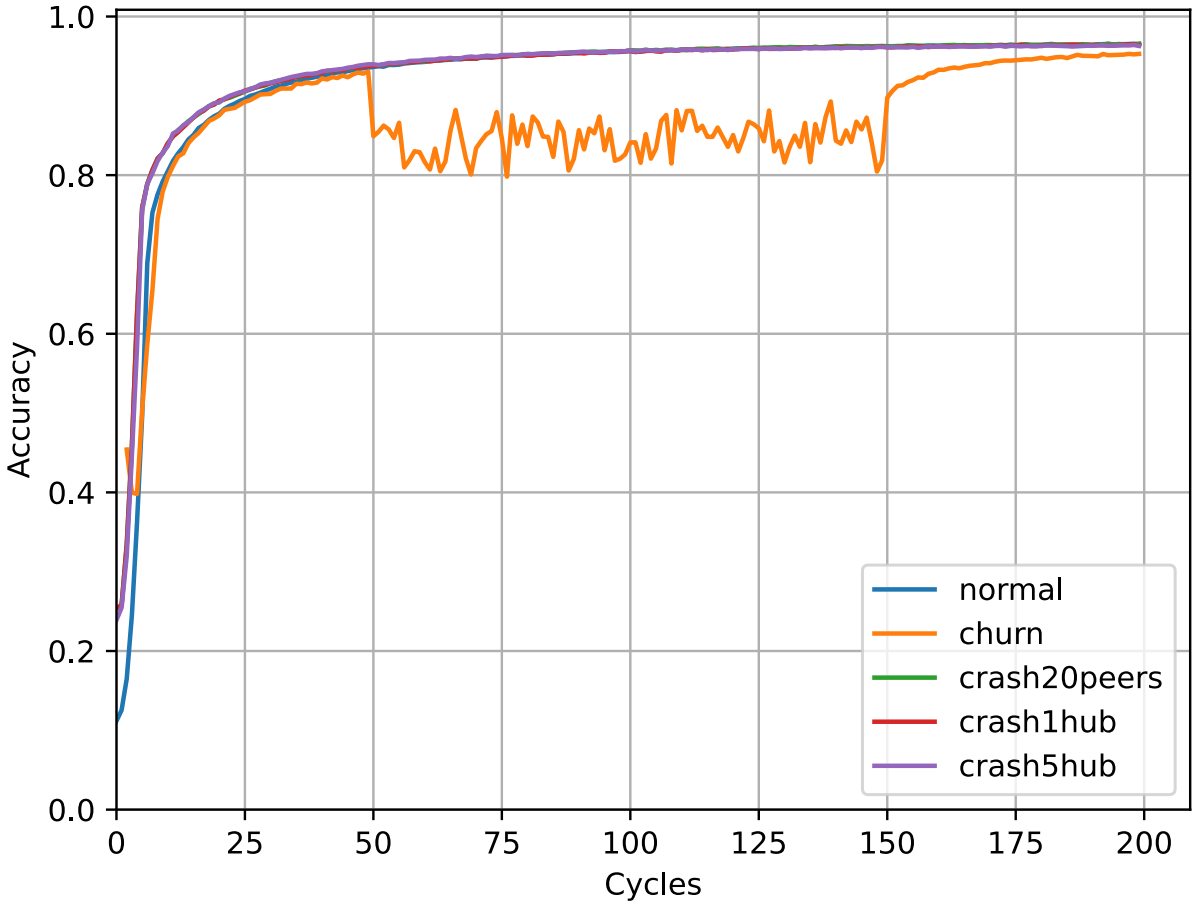
CNN, 60k parameters

Context of experiments

- **Peer-to-Peer**: On the PeerSim simulator.
  - Each generated network had 100 nodes, and the Hub-based topology had 5 hubs
  - In addition to the crash-free context, we ran the algorithm during a context of crash, a context of churn and a context of an attack on the hubs
- **Decentralized Learning**: On the gossipy simulator
  - **ML models**: Logistic Regression & LeNet5
  - **Datasets**: Spambase and MNIST
  - Comparison with Federated Learning, Gossip Learning, Epidemic Learning, GAIA, Chord-based Learning, Fedlay

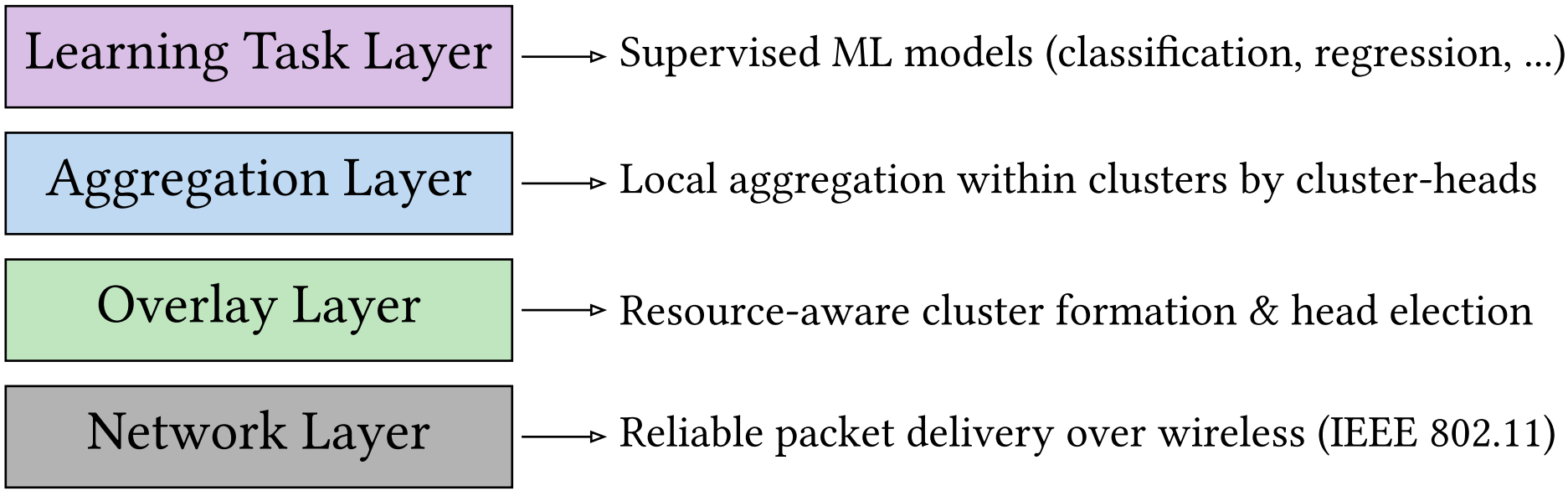

1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0
200
400
600
800
1000
Cycles
Average Accuracy chord
Average Accuracy hub_learning
Average Accuracy epidemic_learning
Average Accuracy federated_learning
Average Accuracy gaia
Average Accuracy gossip_learning
Average Accuracy fedlay
Average Accuracy ring


Accuracy
Cycles
1.0
0.8
0.6
0.4
0.2
0.0
0
250
500
750
1000
1250
1500
1750
2000
1.00
0.95
0.90
0.85
0.80
0
50
100
150
200
h=1, s=1
h=2, s=1
h=3, s=1
h=5, s=1
h=7, s=1
h=9, s=1
h=13, s=1
h=17, s=1
h=21, s=1
h=25, s=1
h=5, s=2
h=7, s=3
h=9, s=4
h=13, s=6
h=17, s=8
h=21, s=10
h=25, s=12


1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0
25
50
75
100
125
150
175
200
Cycles
normal
churn
crash20peers
crash1hub
crash5hub

FLAIR

FLAIR architecture



FLAIR algorithm

FLAIR operates in **rounds** (clustering inspired by LEACH); a fraction $p \in (0, 1)$ of nodes serve as **cluster-heads** (CHs) each round, and the role **rotates** to balance the load.

A node eligible (not a CH during the last $1/p$ rounds) elects itself if $x < T(n)$, with $x \sim \text{Uniform}(0, 1)$ drawn via a **VRF**:

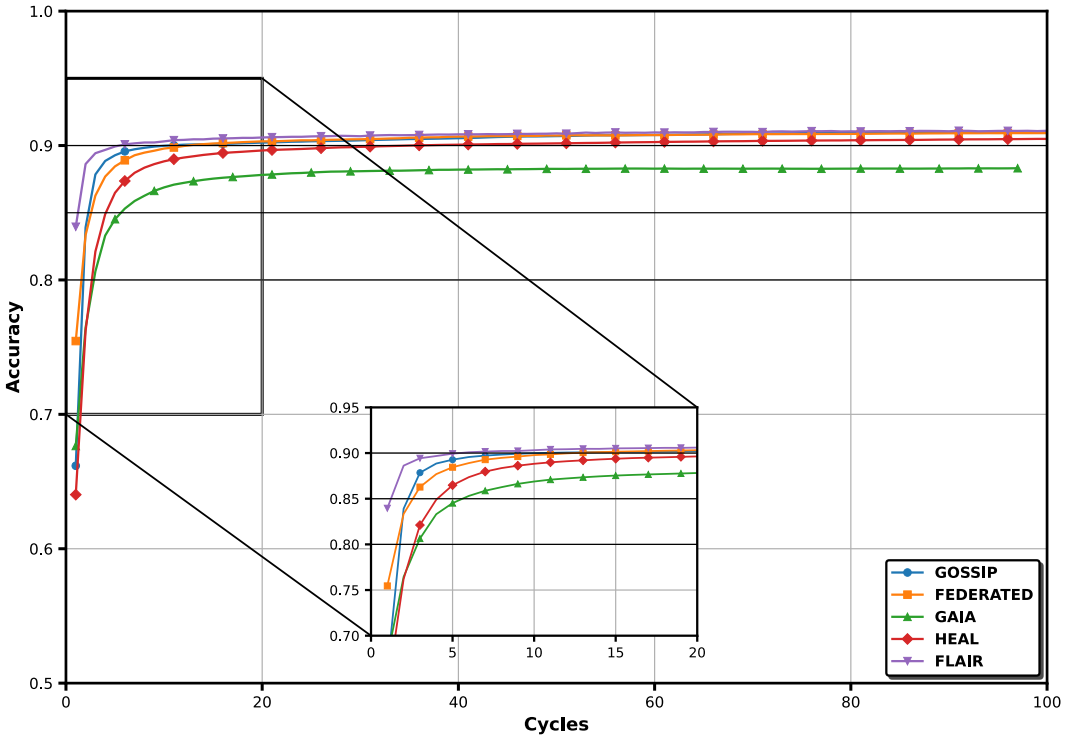
$$T(n) = p \cdot R_n / (1 - p \cdot (r \bmod 1/p))$$

with $R_n = \alpha \cdot \text{CPU}_n + \beta \cdot \text{RAM}_n + \gamma \cdot \text{GPU}_n + \delta \cdot \text{BW}_n$ (resource score; $\alpha + \beta + \gamma + \delta = 1$).

Nodes join the nearest CH; the CH **aggregates locally** within its cluster.

FLAIR simulation results

- Static network (100 nodes): **highest accuracy** ≈ 0.91 (C-FL / HEAL ≈ 0.90, Gossip ≈ 0.88)
- Convergence up to **2.5x faster** than Gaia
- Resilient to **permanent / temporary / random crashes** (up to 90% nodes)
- Tested under **mobility** and in a **smart farming** scenario



Conclusion

Overview

	Decentralized	Fast convergence	Fault-tolerant	Resistant to colluding	Learning	Network type
Elevator	✓	✓	✓	✗	—	P2P
Elevator + Lift	✓	✓	✓	✓	—	P2P
Elevator + HEAL	✓	✓	✓	✗	✓	P2P
FLAIR	✓	✓	✓	✗	✓	Wireless

Limitations

- **Simple ML models.** Benchmarks are modest — logistic regression on Spambase, LeNet5 on MNIST. No large-scale task, so convergence and accuracy conclusions remain to be confirmed on more ambitious models.

- **Combinatorial explosion.** Network parameters (N, c, h), failure/churn scenarios, fraction of malicious nodes, ML hyper-parameters, and seeds define an enormous space — only a small fraction can be explored.

Perspectives — short term

- **Capability-aware hub election** (heterogeneous nodes) and sensitivity to the initial topology.
- Optimise Elevator (message complexity, memory, simplicity).
- Address **data heterogeneity** in HEAL (personalisation, locally-weighted aggregation).
- A unified **Python simulator** for peer sampling + ML.

This work has already begun during a **3-month internship at NII (Tokyo)**.

Perspectives — medium & long term

- Better robustness at the Overlay layer
- **Byzantine robustness in the learning layer** (poisoning & privacy attacks)
- Extend HEAL to **unsupervised and reinforcement learning**.
- **Vertical federated learning** (different features per participant).
- **Formal convergence guarantees** for HEAL.

Publications

- [8] M. A. Legheraba, M. Potop-Butucaru, and S. Tixeuil, "Brief Announcement: A Self-* and Persistent Hub Sampling Service," in *International Symposium on Stabilizing, Safety, and Security of Distributed Systems*, 2024, pp. 461–465.
- [4] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, "Emergent Peer-to-Peer Multi-Hub Topology," in *2024 22nd International Symposium on Network Computing and Applications (NCA)*, 2024, pp. 219–226.
- [9] M. A. Legheraba, N. Rachdi, M. Potop-Butucaru, and S. Tixeuil, "LIFT: Byzantine Resilient Hub-Sampling," in *2025 Thirteenth International Symposium on Computing and Networking (CANDAR)*, 2025, pp. 1–9. — **Outstanding Paper Award**
- [7] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeuil, "HEAL: Resilient and Self-* Hub-based Learning," in *International Conference on Advanced Information Networking and Applications*, 2025, pp. 200–211.
- [10] I. Boutebicha, B. Zaghdoudi, M. A. Legheraba, and M. P. Butucaru, "FLAIR: Distributed Federated Learning with Dynamic Clustering," in *Proceedings of the 14th International Conference on Networked Systems (NETYS 2026)*, in Lecture Notes in Computer Science. Agadir, Morocco: Springer, 2026.
- [11] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, "Des noeuds anonymes hissés au rang de stars," in *ALGOTEL 2025–27èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, 2025.
- [12] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeuil, "Les étoiles montantes de l'apprentissage distribué," in *ALGOTEL 2025–27èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, 2025.
- A journal extension (journal version of this work) has been **submitted to the IEEE/ACM Transactions on Networking**; we are awaiting the **final review** (currently in **minor revision**).

Bibliography

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*, 2017, pp. 1273–1282.

[2] R. Ormándi, I. Hegedűs, and M. Jelasity, "Gossip learning with linear models on fully distributed data," *Concurrency and Computation: Practice and Experience*, vol. 25, no. 4, pp. 556–571, 2013.

[3] I. Hegedűs, G. Danner, and M. Jelasity, "Decentralized learning works: An empirical comparison of gossip learning and federated learning," *Journal of Parallel and Distributed Computing*, vol. 148, pp. 109–124, 2021.

[4] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, "Emergent Peer-to-Peer Multi-Hub Topology," in *2024 22nd International Symposium on Network Computing and Applications (NCA)*, 2024, pp. 219–226.

[5] A.-L. Barabâsi, H. Jeong, Z. Néda, E. Ravasz, A. Schubert, and T. Vicsek, "Evolution of the social network of scientific collaborations," *Physica A: Statistical mechanics and its applications*, vol. 311, no. 3–4, pp. 590–614, 2002.

[6] A. Stavrou, D. Rubenstein, and S. Sahu, "A lightweight, robust p2p system to handle flash crowds," in *10th IEEE International Conference on Network Protocols, 2002. Proceedings.*, 2002, pp. 226–235.

[7] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeuil, "HEAL: Resilient and Self-* Hub-based Learning," in *International Conference on Advanced Information Networking and Applications*, 2025, pp. 200–211.

[8] M. A. Legheraba, M. Potop-Butucaru, and S. Tixeuil, "Brief Announcement: A Self-* and Persistent Hub Sampling Service," in *International Symposium on Stabilizing, Safety, and Security of Distributed Systems*, 2024, pp. 461–465.

[9] M. A. Legheraba, N. Rachdi, M. Potop-Butucaru, and S. Tixeuil, "LIFT: Byzantine Resilient Hub-Sampling," in *2025 Thirteenth International Symposium on Computing and Networking (CANDAR)*, 2025, pp. 1–9.

[10] I. Boutebicha, B. Zaghdoudi, M. A. Legheraba, and M. P. Butucaru, "FLAIR: Distributed Federated Learning with Dynamic Clustering," in *Proceedings of the 14th International Conference on Networked Systems (NETYS 2026)*, in Lecture Notes in Computer Science. Agadir, Morocco: Springer, 2026.

[11] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, "Des noeuds anonymes hissés au rang de stars," in *ALGOTEL 2025–27èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, 2025.

[12] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeuil, "Les étoiles montantes de l'apprentissage distribué," in *ALGOTEL 2025–27èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, 2025.

[13] A. Montresor and M. Jelasity, "PeerSim: A Scalable P2P Simulator," in *Proc. of the 9th Int. Conference on Peer-to-Peer (P2P'09)*, Seattle, WA, Sep. 2009, pp. 99–100.

[14] M. Jelasity, S. Voulgaris, R. Guerraoui, A.-M. Kermarrec, and M. Van Steen, "Gossip-based peer sampling," *ACM Transactions on Computer Systems (TOCS)*, vol. 25, no. 3, p. 8–es, 2007.

[15] J. Huang, L. Kong, G. Chen, Q. Xiang, X. Chen, and X. Liu, "Blockchain-Based Federated Learning: A Systematic Survey," *IEEE Network*, vol. 37, no. 6, pp. 150–157, 2023.

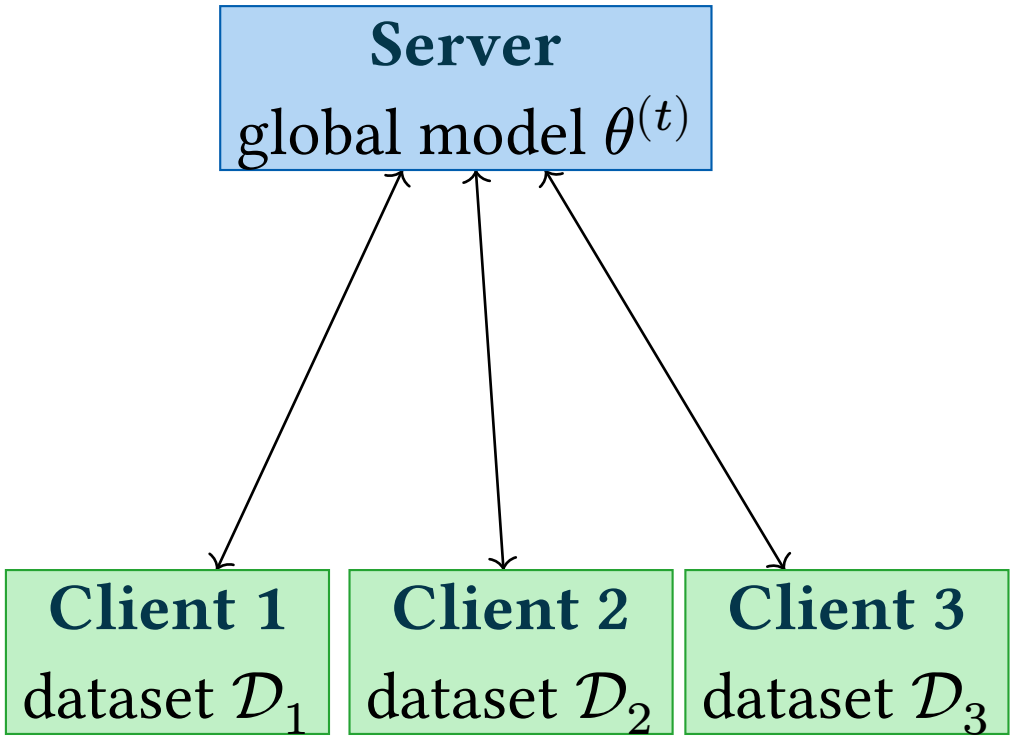
Appendix

Conclusion (detailed)

- **Elevator**: a self-organising overlay that makes hubs emerge — provides a **structured overlay** for structured aggregation without any coordinator.
- **HEAL**: federated learning directly onto the structured overlay — recovers the efficiency of FL while staying decentralised.
- **Lift**: hardens hub election against colluding byzantines (up to 10%).
- **FLAIR**: validates the modular architecture in wireless networks (LEACH-style clustering).

Answer: structured aggregation is possible **without centralisation**, if the overlay layer is designed with that goal in mind.

Federated Learning

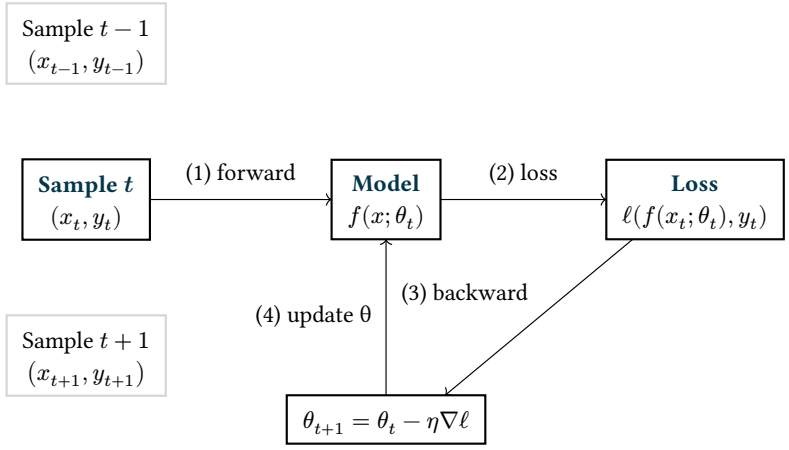


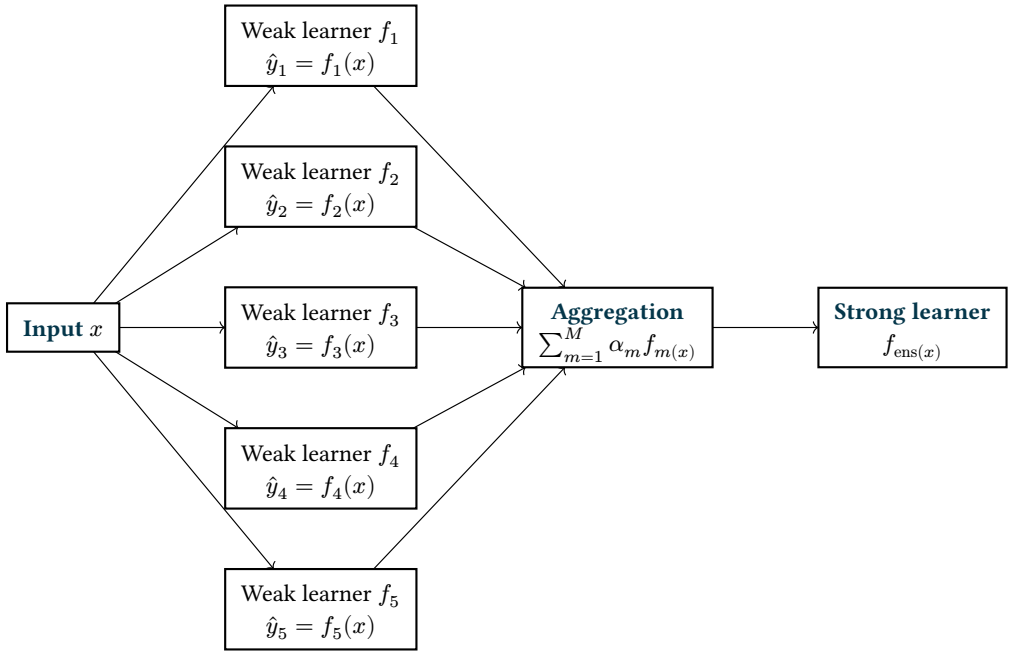
Federated Learning (FedAvg)

for $t = 0$ **to** $T - 1$ **do**
  server broadcasts $\theta^{(t)}$
  for each client i **in parallel**:
    $\theta_i \leftarrow$ local training (lr η, E steps)
  server aggregates: $\theta^{(t+1)} \leftarrow \sum w_i \theta_i$
end for

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*, 2017, pp. 1273–1282.

Federated Learning building blocks





Hub Learning algorithms

HEAL Learning: The Hub Algorithm

```
   duration to wait for model: delta_time
   list of all hubs: hubs_list
1  nb_hubs ← hubs_list.size()
2  loop
3  |  models ← {}
4  |  time ← time.now()
5  |  backwards_list ← {}
6  |  while time.now() < time + delta_time do
7  |  |  (peer_model, peer) ← receive()
8  |  |  models.append(peer_model)
9  |  |  backwards_list.append(peer)
10 |  average_model ← average(models)
11 |  send(hubs_list, average_model)
12 |  models_hubs ← {}
13 |  models_hubs.append(average_model)
14 |  nb_receive ← 0
15 |  while nb_receive < nb_hubs - 1 do
16 |  |  hub_model ← receive()
17 |  |  nb_receive ← nb_receive + 1
18 |  |  models_hubs.append(hub_model)
19 |  global_model ← average(models_hubs)
20 |  send(backwards_list, global_model)
```

SORBONNE UNIVERSITÉ

HEAL Learning: The Client Algorithm

```
   duration to wait for model: delta_time
   The ML model, initialized at random: model
   The local data: data
   list of all hubs: hubs_list
   number of hubs to send the model: s
1  loop
2  |  model ← trainModel(model, data)
3  |  hubs ← chooseRandom(hubs_list, s)
4  |  for hub in hubs do
5  |  |  send(hub, model)
6  |  hubs_models ← {}
7  |  for hub in hubs do
8  |  |  model_hub ← receive()
9  |  |  hubs_models.append(model_hub)
10 |  model ← average(hubs_models)
```

Background thread

Elevator Algorithm (background thread)

```
1 loop
2     request, peer ← receive()
3     if request == CACHE_REQUEST then
4         send(cache, peer)
5         backward_peers.add(peer)
6     if request == BACKWARD_REQUEST then
7         send(backward_peers, peer)
```

Simulation

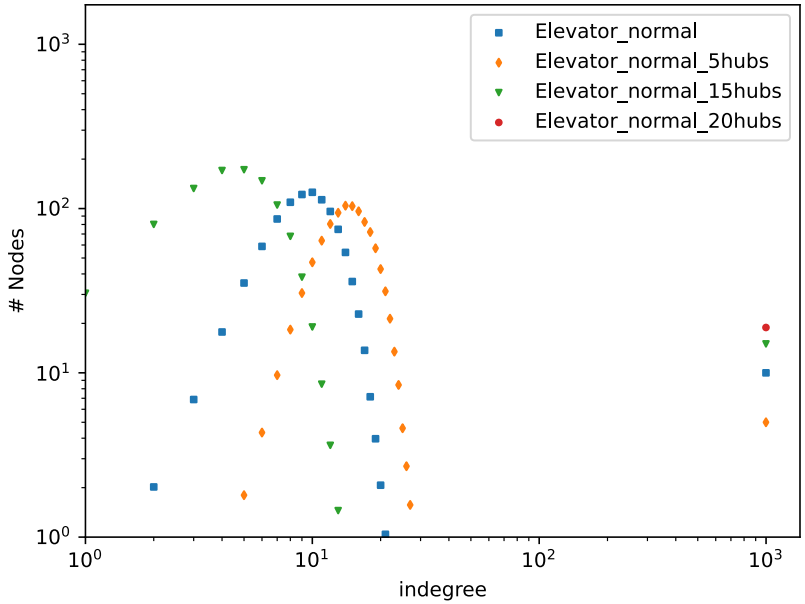
- Simulations done with the Java PeerSim* simulator (modified), with the cycle based mode
- Comparaisons against 3 peer sampling algorithms : Newscast, Proofs and Phenix (Power-law)

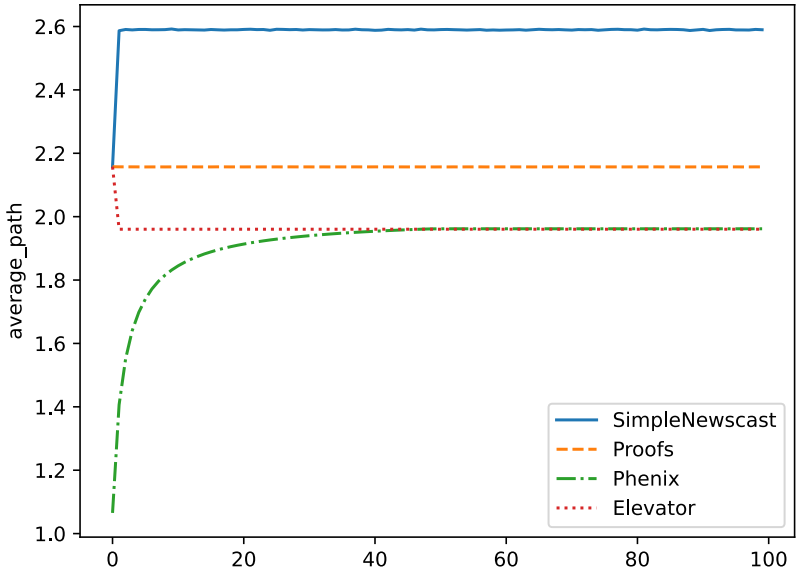
	Value
Network size	1000
Number of cycles for each simulation	1000
Number of times each simulation was run (with different seed)	100
c parameter (cache size)	20
h parameter (number of hubs)	10

- All simulations were run on 16 vCPU, using 64G of memory.
- Code is available at https://gitlab.lip6.fr/legheraba/elevator

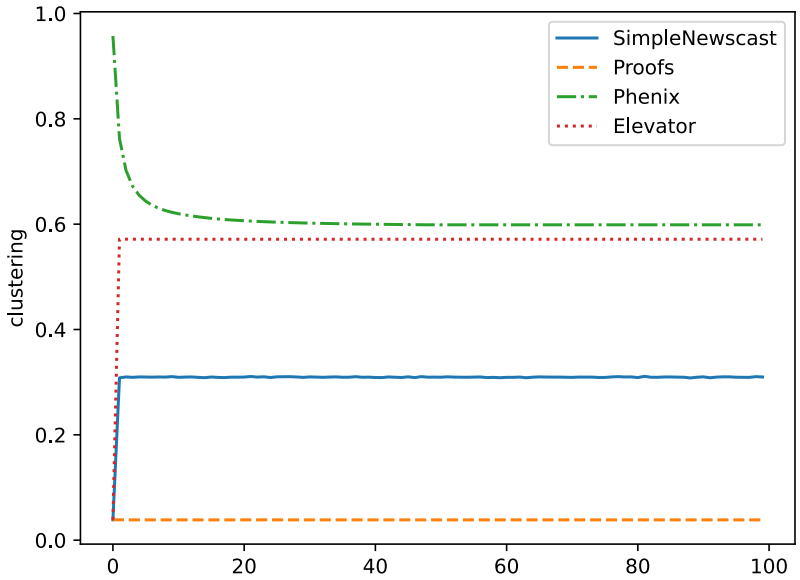
*
[13] A. Montresor and M. Jelasity, "PeerSim: A Scalable P2P Simulator," in *Proc. of the 9th Int. Conference on Peer-to-Peer (P2P'09)*, Seattle, WA, Sep. 2009, pp. 99–100.

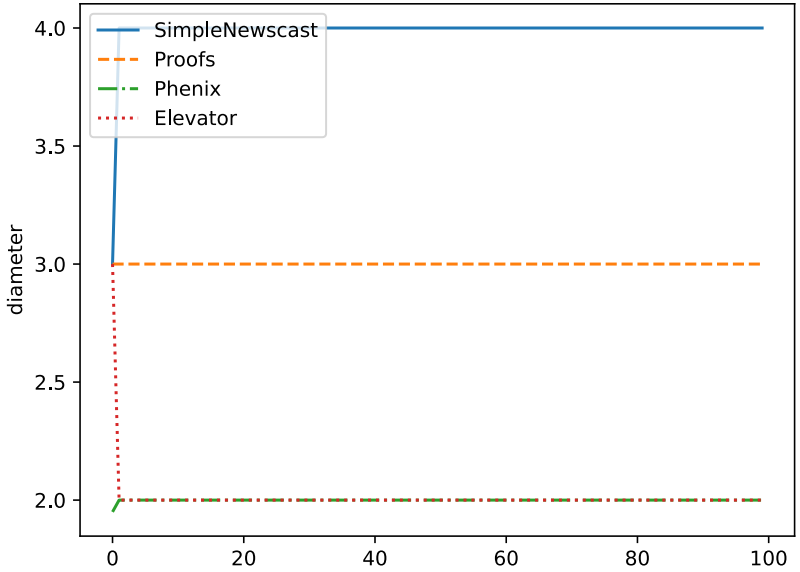
Metrics


Elevator_normal
Elevator_normal_5hubs
Elevator_normal_15hubs
Elevator_normal_20hubs
Nodes
indegree
10^0
10^1
10^2
10^3

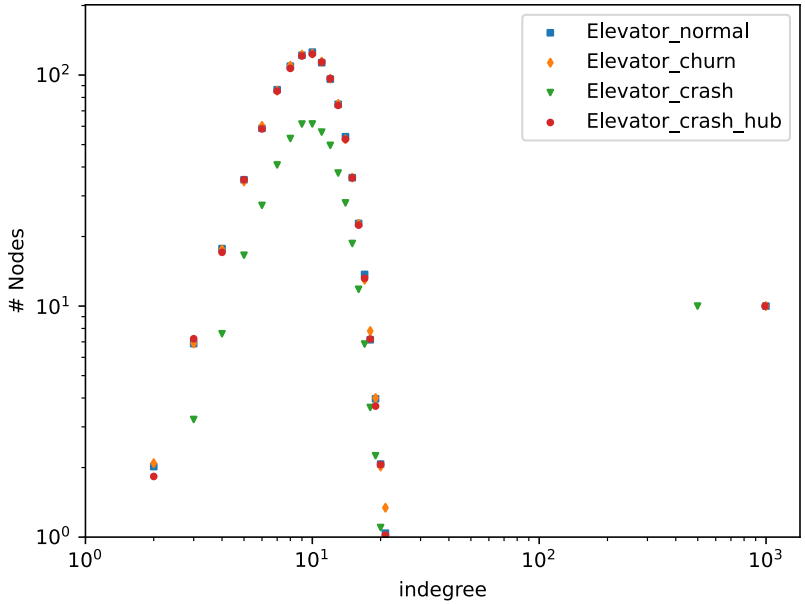

average_path
SimpleNewscast
Proofs
Phenix
Elevator
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
0
20
40
60
80
100

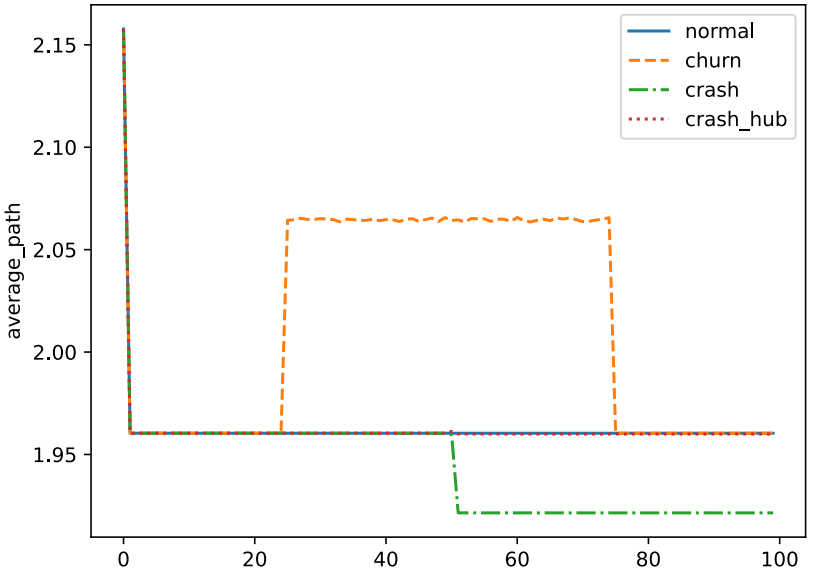
Metrics


clustering
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
SimpleNewscast
Proofs
Phenix
Elevator

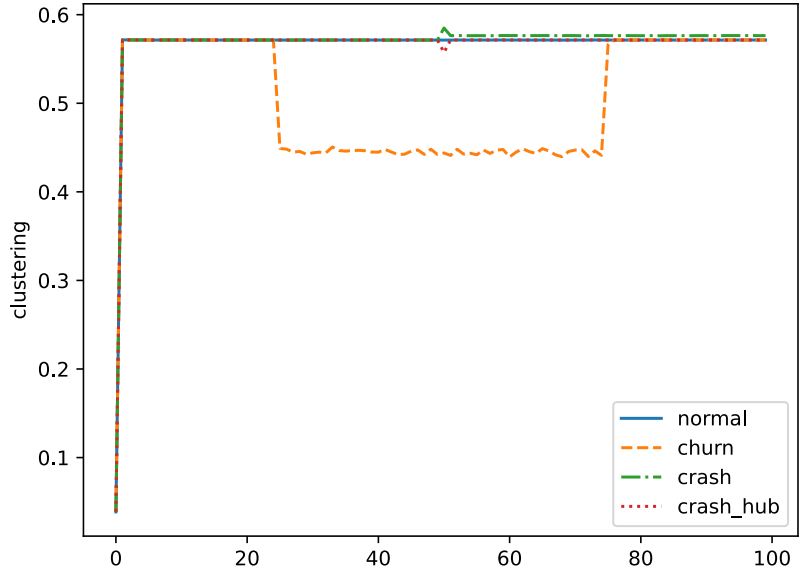

diameter
4.0
3.5
3.0
2.5
2.0
0
20
40
60
80
100
SimpleNewscast
Proofs
Phenix
Elevator

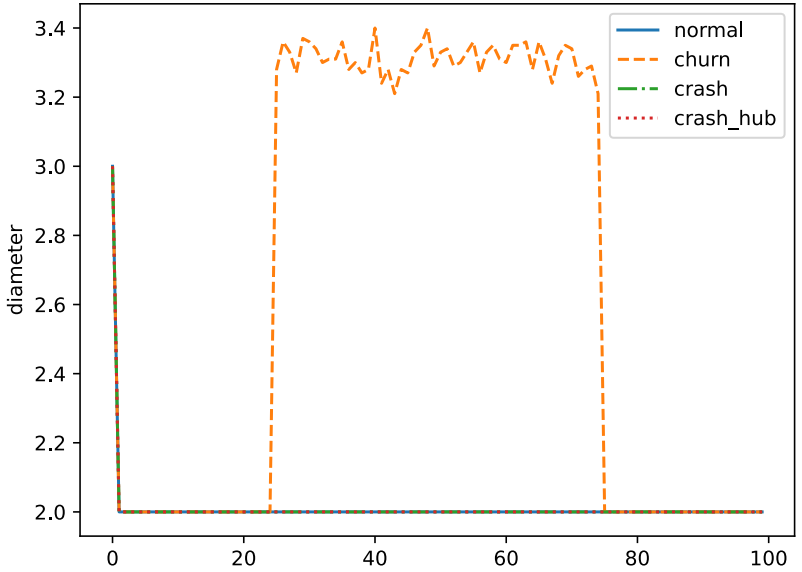
Contexts


Elevator_normal
Elevator_churn
Elevator_crash
Elevator_crash_hub
Nodes
indegree
10^0
10^1
10^2
10^3

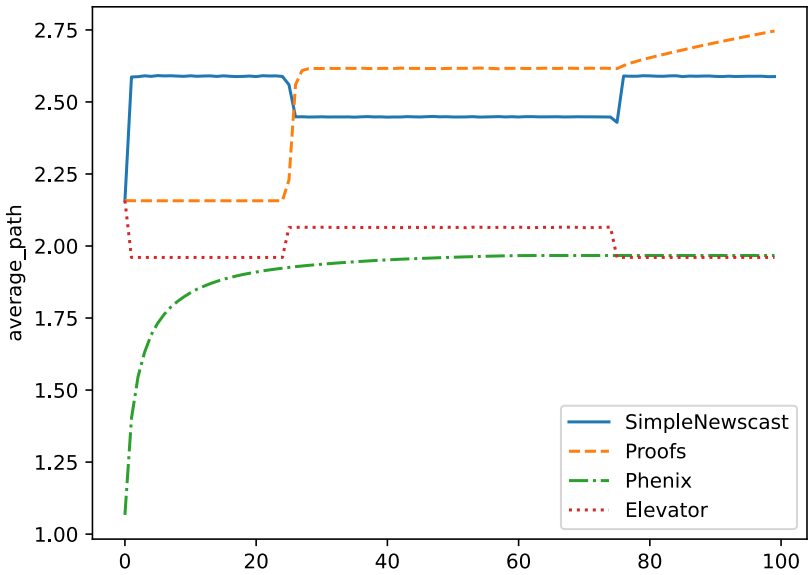

normal
churn
crash
crash_hub
average_path
2.15
2.10
2.05
2.00
1.95
0
20
40
60
80
100

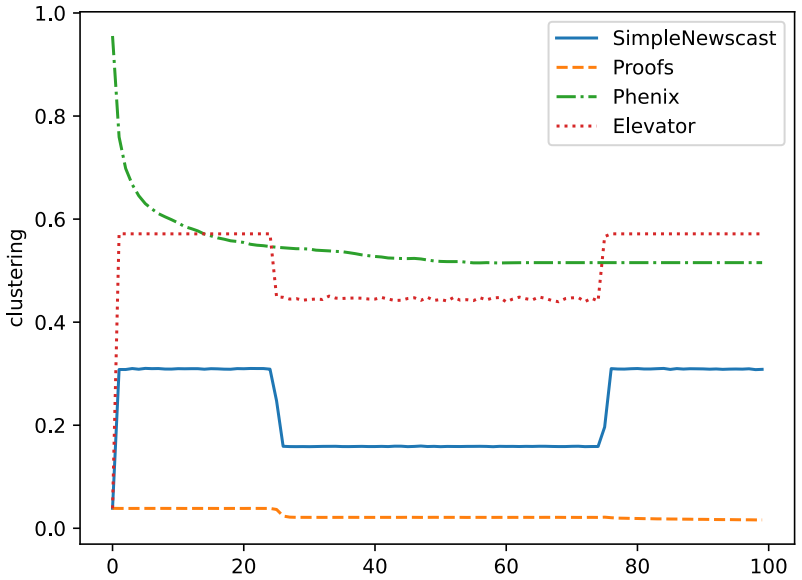
Contexts


clustering
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
100
normal
churn
crash
crash_hub

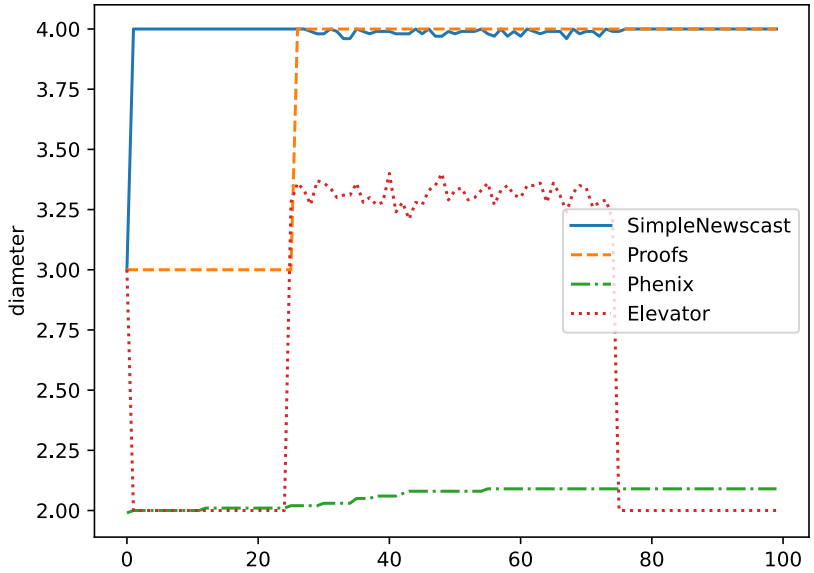

diameter
3.4
3.2
3.0
2.8
2.6
2.4
2.2
2.0
0
20
40
60
80
100
normal
churn
crash
crash_hub

Churn


average_path
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0
20
40
60
80
100
SimpleNewscast
Proofs
Phenix
Elevator


clustering
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
SimpleNewscast
Proofs
Phenix
Elevator

Churn


4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00
diameter
0
20
40
60
80
100
SimpleNewscast
Proofs
Phenix
Elevator

Federated Learning algorithm

Federated Learning (for the Server)

Number of iterations: **T**
Number of clients: **K**
Fraction of clients per iteration: **C**

for each t = 1...T **do**
  m $\leftarrow \max(C \cdot K, 1)$
  $S_t \leftarrow$ selectRandomSubset(m, clients)
  for each $k \in S_t$
    send(k, w_t)
    $w_{t+1}^k \leftarrow$ receive(k)
  $m_t \leftarrow \sum_{k \in S_t} n_k$
  $w_{t+1} \leftarrow \sum_{k \in S_t} \frac{n_k}{m_t} w_{t+1}^k)$

Federated Learning (for a Client)

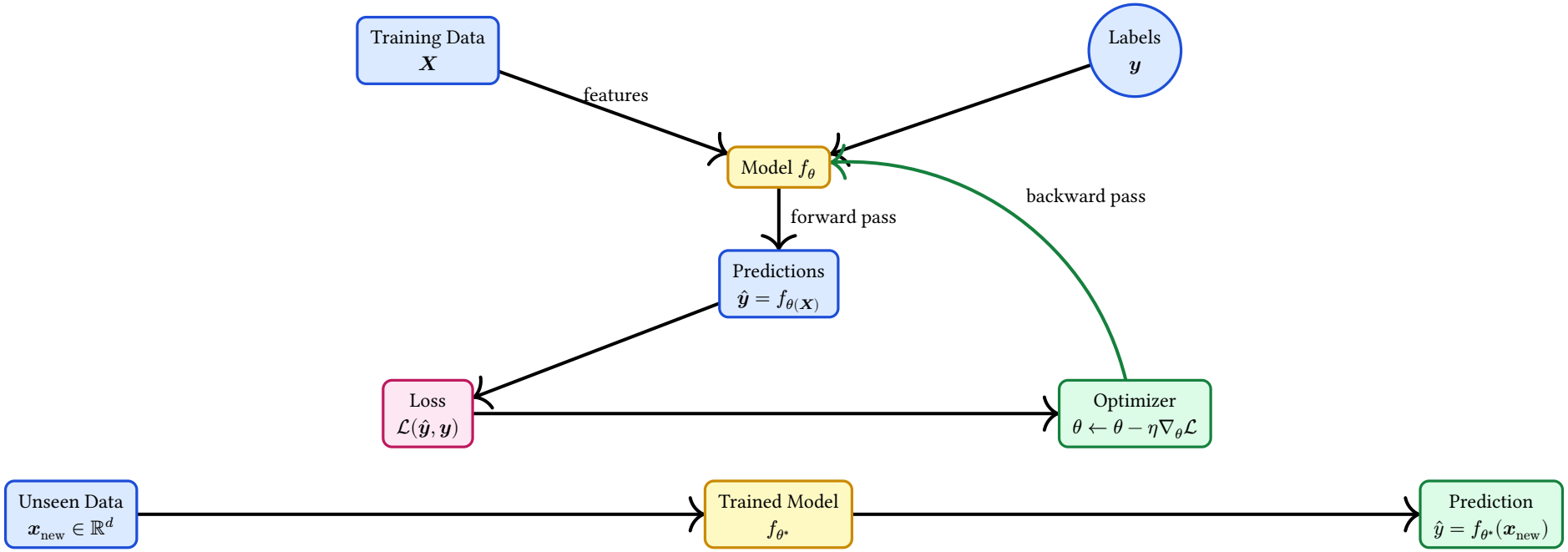
Local dataset: D_k
Number of local epochs: **E**
Learning rate: **η**

receive(w_t)
for each epoch = 1...E **do**
  for batch b $\subset D_k$ **do**
    w_t^karrow.l$w_t - \eta \cdot \nabla L(w_t; b)$
send(w_t^k)

Hub Learning main idea

```
loop
    if node_is_hub do
        while time.now() < time_start + delta_time do
            models_from_nodes.append(receiveModelFromNode())
        aggregated_model ← average(models_from_nodes)
        sendAll(hubs_list, aggregated_model)
        global_model ← average(models_from_other_hubs)
        send(nodes_list, global_model)
    else
        hubs ← chooseRandom(hubs_list, number_hubs_to_send_model)
        model ← trainModel(model, data)
        sendAll(hubs, model)
        for hub in hubs do
            global_models.append(receiveGlobalModelFromHub())
        model ← average(global_models)
```

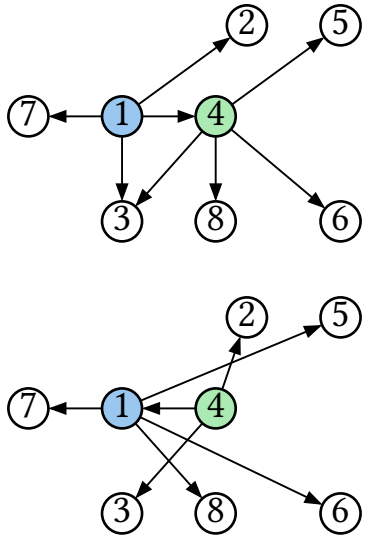
Machine Learning



Decentralized peer sampling

PROOFS Algorithm (active thread)

```
   initial peer list: cache
   cache size: c
   shuffle length: l
   node address: p
1  loop
2      wait(Δ)
3      subset ← selectRandomSubset(cache, l)
4      q ← selectRandom(subset)
5      subset.remove(q)
6      subset.add(p)
7      send(q, subset)
8      subset_q ← receive(q)
9      subset_q.remove(p)
10     subset_q.removeAll(cache)
11     cache ← subset_q
```



Before and after a shuffling operation. Node 1 sends addresses {itself, 2, 3} to node 4. Node 4 sends back {5,6, 8}.

[6] A. Stavrou, D. Rubenstein, and S. Sahu, "A lightweight, robust p2p system to handle flash crowds," in *10th IEEE International Conference on Network Protocols, 2002. Proceedings.*, 2002, pp. 226–235.

Hub-based topology & Hub sampling

- The objective is to obtains an overlay network with **h** defined hubs, with **h** a parameter of the algorithm, and each hub is connected to all the nodes in the networks. The application running on top will be able to take advantage of this overlay to speed up message transmission in the network.
- **Preferential Attachment**: Drawing from the concept pioneered by Barabási and Albert[*], preferential attachment dictates that new connections in the network are established preferentially with nodes possessing a higher number of existing connections.
- **Random Attachment**: Inspired by gossip-based peer sampling algorithms ([**]), random attachment ensures that nodes maintain connections with a representative and diverse subset of the network.

[4] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, "Emergent Peer-to-Peer Multi-Hub Topology," in *2024 22nd International Symposium on Network Computing and Applications (NCA)*, 2024, pp. 219–226.

---

* [5] A.-L. Barabâsi, H. Jeong, Z. Néda, E. Ravasz, A. Schubert, and T. Vicsek, "Evolution of the social network of scientific collaborations," *Physica A: Statistical mechanics and its applications*, vol. 311, no. 3–4, pp. 590–614, 2002.

* [6] A. Stavrou, D. Rubenstein, and S. Sahu, "A lightweight, robust p2p system to handle flash crowds," in *10th IEEE International Conference on Network Protocols, 2002. Proceedings.*, 2002, pp. 226–235.

* [14] M. Jelasity, S. Voulgaris, R. Guerraoui, A.-M. Kermarrec, and M. Van Steen, "Gossip-based peer sampling," *ACM Transactions on Computer Systems (TOCS)*, vol. 25, no. 3, p. 8–es, 2007.

Elevator's Algorithm

Elevator Algorithm (active thread)

```
   initial peer list: cache
   cache size: c
   desired number of hubs: h
   initial backward list: backward_peers (empty)
1  loop
2      wait(Δ)
3      frequency_map ← {}
4      for peer in cache do
5          peer_cache ← send(CACHE_REQUEST, peer)
6          frequency_map ← frequency_map ∪ peer_cache
7      preferred ← frequency_map.sort().select(h)
8      preferred_backward ← {}
9      for peer in preferred do
10         peer_backward_peer ← send(BACKWARD_REQUEST,
           peer)
11         preferred_backward.add(peer_backward_peer)
12     cache ← {}
13     cache ← preferred ∪ preferred_backward
14     backward_peers ← {}
```

SORBONNE UNIVERSITÉ

Elevator Algorithm (background thread)

```
1  loop
2      request, peer ← receive()
3      if request == CACHE_REQUEST then
4          send(cache, peer)
5          backward_peers.add(peer)
6      if request == BACKWARD_REQUEST
       then
7          send(randomValue(backward_peers),
           peer)
```

Blockchain-based Federated Learning

[15] J. Huang, L. Kong, G. Chen, Q. Xiang, X. Chen, and X. Liu, "Blockchain-Based Federated Learning: A Systematic Survey," *IEEE Network*, vol. 37, no. 6, pp. 150–157, 2023.

Services in a P2P Network

SORBONNE UNIVERSITÉ

Peer sampling

Peer discovery

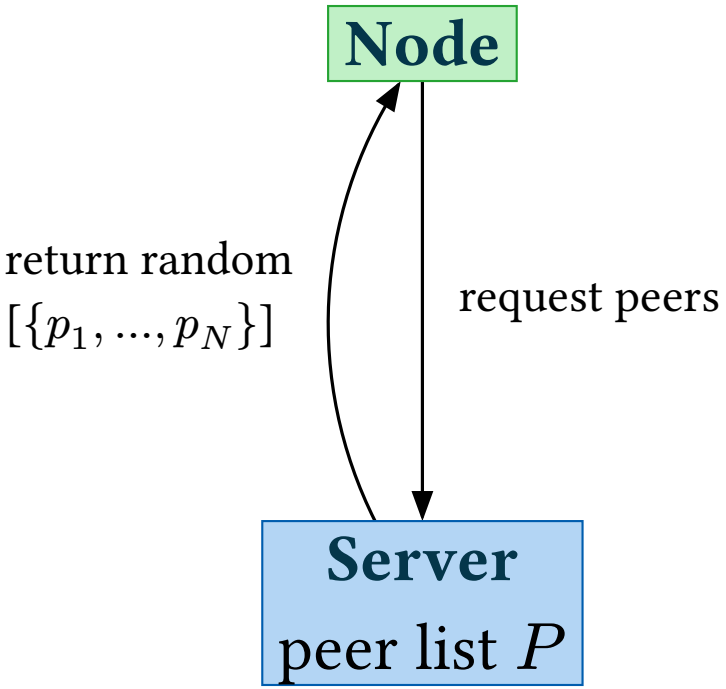
Data request

Membership management

Topology management

Information dissemination

Peer sampling



Definition. (Partial View) The **partial view** of v, denoted $P(v)$, is the set of nodes v can send messages to, $P(v) \subseteq V$, with $|P(v)| \leq c$, $c \ll N$, and $P(v) = \text{neigh}_G(v)$.